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Counter-clockwise rotations in the southern Apennines during the Pleistocene: paleomagnetic evidence from the Matera area

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ABSTRACT

Sediments from late Pliocene to early Pleistocene age have been sampled at six different localities/sections (38 sites) in the Matera area which is in the southern part of peninsular Italy (Basilicata). Three of the localities are from the Apenninic belt, the other three from the Bradano Unit, both of them west of the earlier sampled Apulian Unit.

Stepwise thermal demagnetisation revealed that usually two important components contribute to the natural remanent magnetisation (NRM) of these clay sediments: a low-temperature (LT) component removed between 100 and 300–320°C and a high-temperature (HT) component removed between 300–320 and 430–450°C. The HT component is a characteristic remanent magnetisation (ChRM) for the sediments and has a clear primary (= pre-folding) origin. In fresh, unweathered sediments the LT component is identical in direction to the HT component, but there is evidence that moderate weathering replaced a high-intensity primary LT component with a low-intensity secondary LT magnetisation. The HT magnetisation is affected by this weathering process in a decrease of its intensity.

A counter-clockwise deflection in declination is observed for the sites sampled in the Apenninic belt. The HT directions reveal a 23° counter-clockwise rotation relative to the magnetic north: $D_m = 336.6°$ and $I_m = 57.2°$. Sites in the Bradano Unit have no tectonic rotation: $D_m = 359.1°$ and $I_m = 56.3°$. Earlier sampled sites from the Apulian Unit also showed no tectonic rotation: $D_m = 359.4°$ and $I_m = 57.0°$. Paleomagnetic data support the hypothesis that the southern Apennines rotated in the middle–late Pleistocene, after the major phase of extension in the southern part of the Tyrrhenian basin and contemporaneous with a regional compressional phase.

Introduction

The southern part of peninsular Italy shows a complex pattern of tectonic activities during the Neogene and Quaternary (see, e.g., Patacca and Scandone, 1989). There is mountain building for the central part of the area, extension in the internal part. Depocentre shifting and block rotations are direct manifestations of tectonic activity. The southern Apennines were created during the Neogene and Quaternary; tectonic activity and sedimentation shifted eastwards and southwards; the present foredeep is the Gulf of Taranto (Pescatore and Senatore, 1986). An important mechanism that has caused these tectonic activities is the passive subduction of a lithospheric slab by gravitational sinking. Possibly, tectonic block rotations have occurred during this process, but they are not very well constrained, temporally nor geographically. From a paleomagnetic point of view, little has been done so far to resolve the timing and geographical distribution of the possible block rotations of this area during the Neogene and Quaternary. Only recently, the first paleomagnetic studies (Sagnotti, 1992; Scheepers, 1992) appeared, from Pliocene and Pleistocene sediments, and showed that considerable rotations indeed occurred.

We have sampled sediments from the late Tertiary and Quaternary in the Matera area (Basilicata province, Fig. 1) for paleomagnetic purposes with the aim of extending the paleomagnetic work done thus far.

Rocks of pre-Neogene age have been studied extensively from the southern Apennines and its foreland, in particular Cretaceous limestones from the Apulian foreland (Channell and Tarling, 1975; Channell, 1977; VandenBerg, 1983). The results of these workers revealed a characteristic mean direction for these limestones that implies

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a counter-clockwise rotation of approximately 30° since the early Tertiary. Mesozoic limestones from the southern Apennines reveal counter-clockwise rotations between 34 and 112° (Channell and Tarling, 1975; Catalano et al., 1976; Incoronato, 1983; K.C. Jackson, 1990). The work of Sagnotti (1992) established a 22° counter-clockwise rotation from early Pleistocene clays from the Sant'Arcangelo basin, whereas early Pleistocene clays from the Apulia–Gargano foreland show no rotation (Scheepers, 1992). All these tectonic rotations are of great interest, but many more studies are needed to separate local and regional effects and to determine the age and causes of the rotations. This paper is a contribution to this: it reports the study of early Pleistocene sediments from different tectonic settings in the Matera area. The results are compared with the earlier established paleomagnetic data in this area.

Geological setting

From a purely geological point of view, Southern Italy is divided into three main tectonic units: the Apenninic belt, the Bradano Unit and the Apulia–Gargano foreland. The most easterly and external one, the Apulia–Gargano foreland (Fig. 1), has a continental basement covered by a carbonate succession of Mesozoic to early Tertiary age that exceeds 6000 m in thickness and is called the Apulia Unit (D'Argenio et al., 1975). This Apulia Unit extends below the deposits of the Bradano Unit (Fig. 1) which are Plio-Pleistocene in age. The foreland is further characterised by a crustal thickness of about 30 km, whereas the crustal thickness below the southern Apennines can be as much as 35–40 km. The foredeep basin was considered to have a flexural origin and is the result of the retreat of the subducting lithospheric slab towards the foreland (Royden et al., 1987).

The Bradano Unit, more to the west, is a terrigeneous succession, several thousands of metres thick that was accumulated in a Plio-Pleistocene foredeep (Pescatore and Senatore, 1986). These young sediments as well as the underlying Mesozoic–early Tertiary carbonate sequence are affected by normal faulting (Cristofolini et al., 1985). Compressive structures are only detectable in the western part, in connection with the Apen-

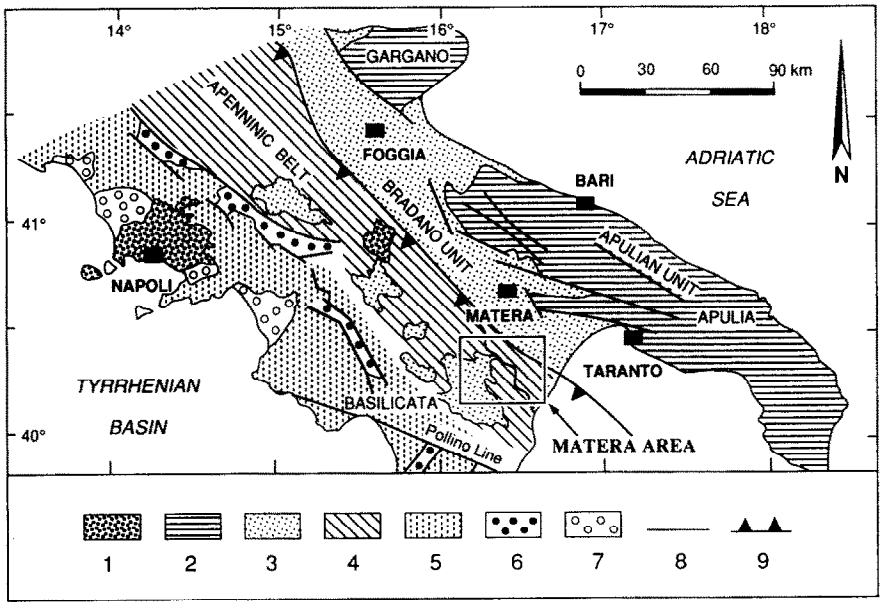


Fig. 1. Major tectonic units in the Southern Apennines and the locality of the Matera area (modified after Albarello et al. 1990; Cristofolini et al. 1983). *1* = volcanic rocks; *2* = Apulia–Gargano foreland *3* = Pliocene–Pleistocene deposits; *4* = external Apenninic belt; *5* = internal Apenninic belt; *6* = internal grabens; *7* = internal "Peri-Tyrrhenian" deposits; *8* = faults; *9* = Apenninic front.

ninic belt system. Most of the 320 km of shortening in the southern Apennines took place between the Langhian and the Tortonian stages of the Miocene (D'Argenio et al., 1975).

Sampling and laboratory treatment

Sampling sites were mainly selected on the basis of sedimentary/lithological criteria. Fine-grained sediments (marine marls and clays) were chosen because of their suitable magnetic properties and because their sedimentation rate is usually low enough to average out secular variation, to a large extent. Sections and/or outcrops close to major faults or containing landslides were avoided.

The present study deals with late Tertiary and Quaternary marly clays from the Bradano Unit and the Apenninic belt in the Matera area. The colour of the sampled clays is an indication of the freshness of the clays: fresh clays are mostly blue coloured and give better paleomagnetic results. Sometimes it was necessary to remove the weathered surface before drilling. A total of 235 cores from 38 sites was sampled. We have divided the sampling into six different localities (Fig. 2). We took cylindrical standard cores of 25 mm diameter and 6–10 cm length using an electric drill and portable generator.

Three of these localities belong to the Apenninic belt: (1) an area in the northern part of the Sant'Arcangelo basin; (2) an area in the southern part of the Sant'Arcangelo basin; and (3) the Craco area. The variety in bedding tilts for the sites in these three areas enables to test for the primary (= pre-tilt) origin of the remanent magnetisations. The other three localities belong to the Bradano Unit (Fig. 2): (1) the Tursi area

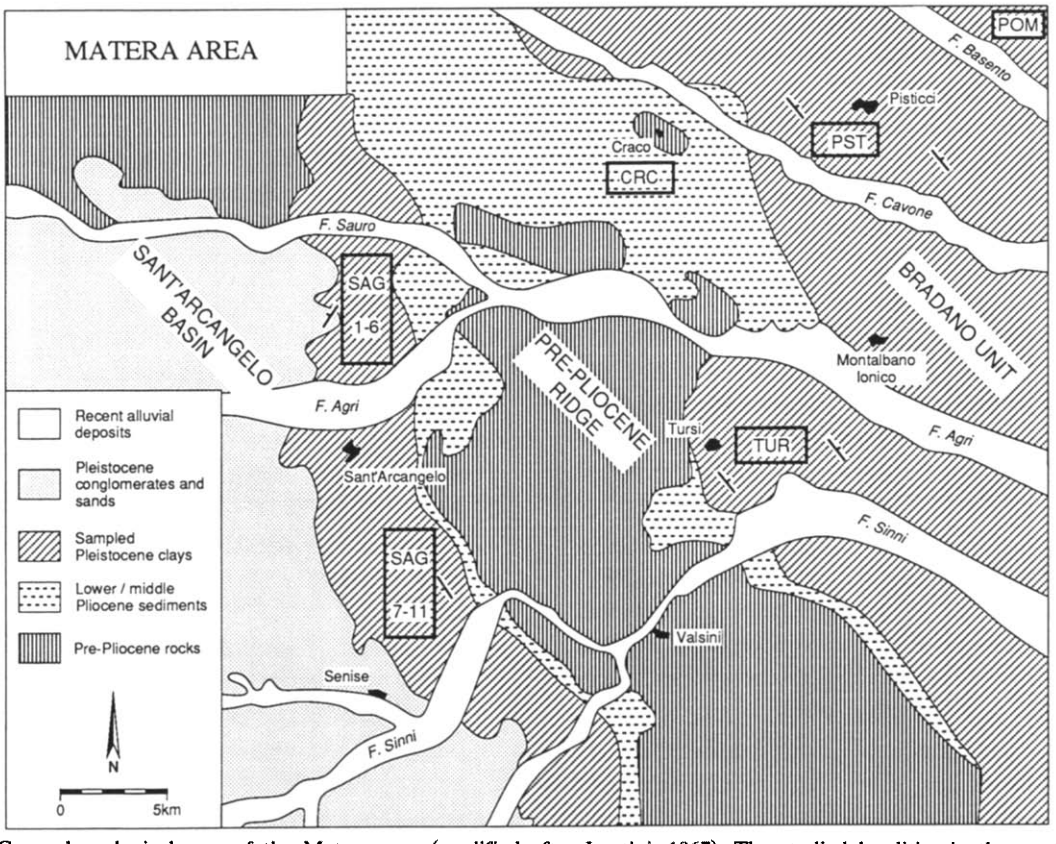


Fig. 2. General geological map of the Matera area (modified after Lentini, 1967). The studied localities in the western part (Apenninic belt) are the two localities in the Sant'Arcagelo basin (*SAG 1–6*, *SAG 7–11*) and the Craco locality; in the eastern part the three localities are in the Bradano Unit (Pisticci, Pomárico and Tursi).

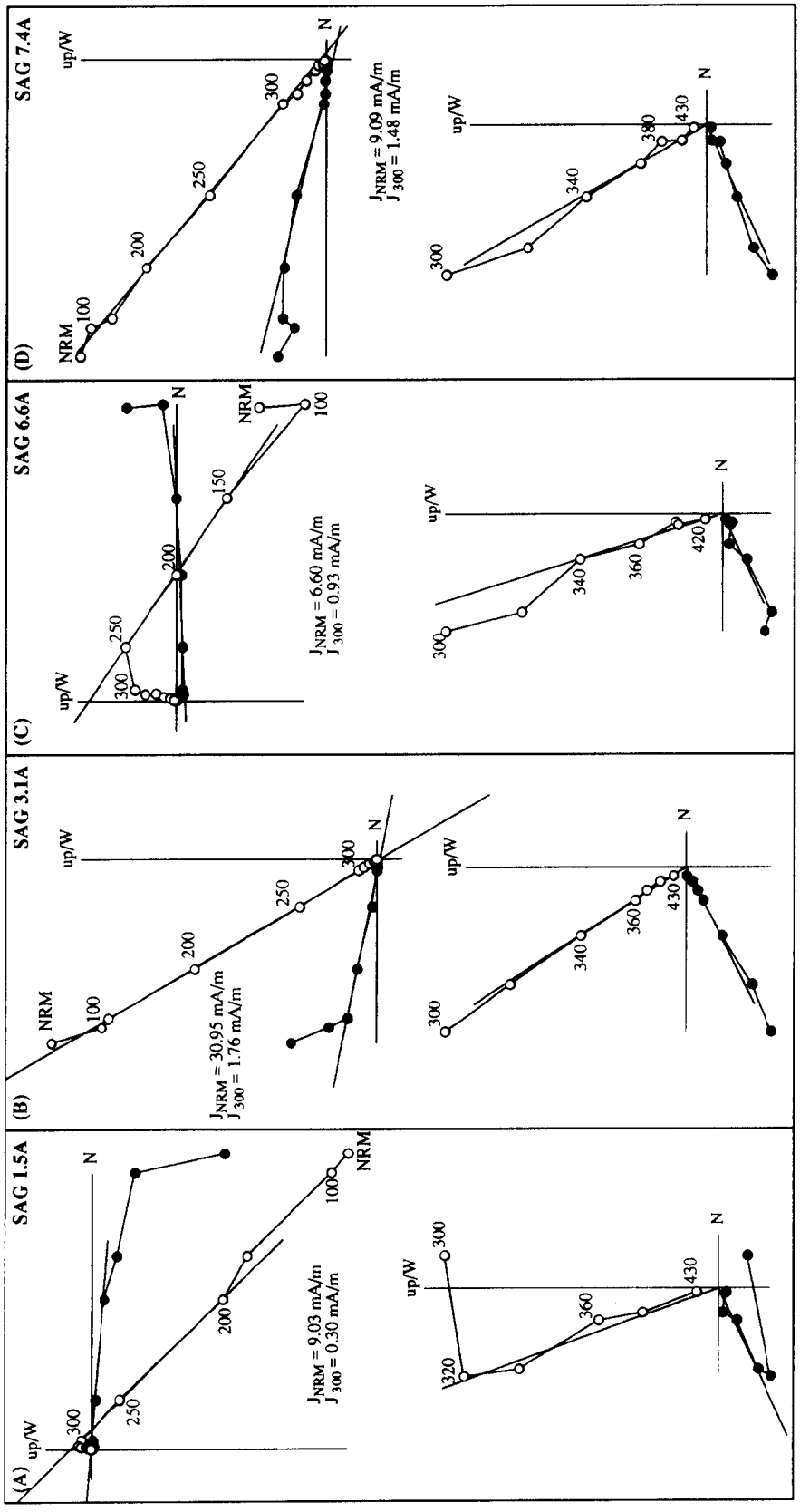


Fig. 3. Stepwise thermal demagnetisation diagrams of samples from the Sant'Arcangelo basin (*SAG*) area. The upper diagrams are given without bedding tilt correction and represent the thermal demagnetisation of the low-temperature (LT) component. Lower diagrams give a blow-up of the upper diagrams and include bedding tilt correction; they characterize the demagnetisation of the high-temperature (HT) component. Numbers denote demagnetisation steps in °C. Direction and intensity of magnetisation during thermal demagnetisation were analysed by vector diagram representation (Zijderveld, 1967) and principal component analysis (Kirschvink, 1980).

between the Sinni and Agri rivers; (2) the Pisticci area north of the Tursi area between the Cavone and Basento rivers; and (3) the Pomárico area north of Pisticci between the Basento and Brádano rivers. The sites in the Bradano Unit are mostly in moderately tilted sequences.

Paleomagnetic results of the Matera area

Introduction

In the laboratory, routine paleomagnetic methods were applied to obtain characteristic magnetisation directions for the samples. The natural remanent magnetisations (NRM) were measured partly with a JR-3 spinner magnetometer with a digital interface and partly with a 2G cryogenic magnetometer. The initial NRM intensities of these sediments range from 0.05 to 70 mA/m. The NRM of generally six specimens per site was analysed by means of progressive stepwise thermal demagnetisation using small temperature increments (50–20°C). The low-field bulk susceptibility (X_{LF}) of the samples during thermal treatment was measured on a Bartington MS2 susceptometer, to detect changes in the magnetic mineralogy. Maximum demagnetisation temperatures were determined by the limit of reproducible results. This limit was strongly influenced by a large increase in X_{LF} and viscous behaviour at temperatures higher than 400–450°C. The viscous behaviour relative to the remanent magnetisation then becomes too large to measure a meaningful remanence direction.

In this paper, firstly the general NRM characteristics of the sediments are reported, and secondly, some rock magnetic results will be discussed. Demagnetisation diagrams (Zijderveld, 1967) were used for the interpretation of characteristic remanence magnetisation directions. For low-intensity sites, stacked demagnetisation diagrams (Scheepers and Zijderveld, 1992) were calculated. The characteristic directions of the magnetisations were determined by least squares fitting of lines through selected datapoints (Kirschvink, 1980). From the characteristic remanent magnetisation (ChRM) directions a site mean direction is calculated and then from the site means a locality mean using Fisher (1953) statistics.

The Matera area, Apenninic belt

The Sant'Arcangelo basin

In the Sant'Arcangelo basin, all sites were collected along the eastern basin margin (Fig. 2) in clays of late Pliocene to early Pleistocene age (formation P^3Q—foglio 211, Carta geologica d'Italia). This eastern basin margin was also sampled by Sagnotti (1992); his results indicate that the Sant'Arcangelo (piggy-back) basin underwent a counter-clockwise rotation of 22°. In our sampling, sites SAG 1–6 are situated north of the village Sant'Arcangelo between the Agri and Sauro rivers (Fig. 2) in an area east of Aliano. Most sites in this northern area were not optimally fresh, as was indicated by some brown spots in the mostly blue-coloured sediments; only site SAG 3 was in very fresh clays (completely blue coloured with dark dots). The bedding tilt in the northern area is between 20 and 40° to the north, west and northwest (Table A2). In the area south of the village of Sant'Arcangelo, sites SAG 7–11 were all drilled in optimally fresh outcrops along the new road (SS92) to Senise in the valley of the Sant'Arcangelo river. In this southern area the bedding tilt for the outcrops is between 18 and 30° to the southwest (Table A2). The difference in general bedding tilt between the northern and southern area of the Sant'Arcangelo basin makes it possible to perform a meaningful fold test. The marly clays of the Sant'Arcangelo basin are overlain by Pleistocene sands and conglomerates belonging to the Aliano formation (see foglio 211, Carta geologica d'Italia). According to our own biostratigraphic age estimates (see Appendix), the Sant'Arcangelo sites most probably have an age younger than 1.35 Ma, although a slightly older age (with a maximum of 2.02 Ma) cannot be excluded for a number of sites (SAG 1, 4, 7, 8 and 9); the stratigraphical order of the sites could not (reliably) be determined.

The paleomagnetic results from the sediments belonging to the Sant'Arcangelo basin show two well-defined NRM components (Fig. 3), apart from a small viscous (laboratory induced) compo-

TABLE 1

Site mean directions and overall mean directions from the sites of the Apenninc belt sampled in the Matera area using Fisher (1953) statistics

Site	N	D	I	D(tc)	I(tc)	k	α_{95}	Int	Comp
Sant'Arcangelo area (north)									
SAG 1	6	13.1	42.0	8.3	7.3	132	5.8	5.57	LT
SAG 2	5	253.0	−19.6	238.2	−58.4	52	10.8	0.19	LT
SAG 3	5	184.6	−62.1	150.2	−57.5	357	4.1	17.64	LT
SAG 4	5	24.5	39.3	25.7	9.4	218	5.2	1.48	LT
SAG 5	5	359.6	38.3	352.7	21.8	98	7.8	1.90	LT
SAG 6	5	356.9	38.0	357.6	0.0	117	7.1	4.32	LT
SAG 1	6	37.2	−75.6	152.7	−63.0	24	14.0	0.16	HT
SAG 2	5	216.6	−52.2	152.4	−56.3	309	4.4	0.22	HT
SAG 3	5	176.7	−59.5	147.6	−53.1	398	3.8	1.03	HT
SAG 4	5	91.9	−55.3	139.7	−57.8	44	11.6	0.25	HT
SAG 5	5	95.2	−88.9	137.8	−69.2	179	5.7	0.31	HT
SAG 6	6	42.6	−75.1	158.7	−61.4	195	4.8	0.61	HT
Sant'Arcangelo area (south)									
SAG 7	5	183.7	−48.4	151.9	−60.6	56	10.4	5.73	LT
SAG 8	5	181.1	−40.3	157.5	−52.9	51	10.8	1.33	LT
SAG 9	3	174.7	−26.1	156.8	−38.6	8	47.4	1.81	LT
SAG 10	6	183.6	−38.4	174.4	−54.2	21	15.0	9.49	LT
SAG 11	6	176.5	−29.2	168.2	−51.3	1181	2.0	29.63	LT
SAG 7	5	178.8	−49.2	145.7	−59.2	87	8.2	1.70	HT
SAG 8	5	182.3	−45.1	154.1	−57.4	648	3.0	0.85	HT
SAG 9	5	190.0	−34.5	166.2	−52.8	375	4.0	0.37	HT
SAG 10	6	187.2	−42.6	177.6	−58.9	251	4.2	2.43	HT
SAG 11	6	179.0	−32.7	170.4	−55.2	873	2.3	6.39	HT
Craco area									
CRC 1	6	359.7	44.8	5.9	22.5	17	16.7	0.70	LT
CRC 3	6	345.2	73.0	354.8	51.5	11	21.3	1.68	LT
CRC 4	6	342.3	41.0	322.7	50.6	51	9.5	0.10	LT
CRC 5	5	334.8	66.3	349.9	29.1	47	11.3	1.97	LT
CRC 1	2	146.2	−56.0	166.5	−39.8	111	24.0	0.60	HT
CRC 2	7	104.3	−68.8	156.8	−61.5	214	4.1	0.51	HT
CRC 3	6	310.3	72.0	338.1	54.3	192	4.8	1.55	HT
CRC 4	5	191.5	−43.7	176.8	−61.8	78	8.7	0.15	HT
CRC 5	5	243.6	64.0	318.2	56.1	86	8.3	0.63	HT
CRC 6	4	176.8	−47.9	149.9	−45.1	8	35.1	0.05	HT
Mean directions: Sant'Arcangelo basin									
4 sites (N)	no tc			8.4	39.9	65	11.5	3.32	LT
	tc			6.2	9.8	23	19.6	3.32	LT
7 sites (R)	no tc			190.9	−40.9	9	20.9	9.40	LT
	tc			168.0	−56.3	21	13.6	9.40	LT
6 sites (R)	no tc			137.6	−81.0	10	22.9	0.43	HT
north	tc			148.5	−60.3	140	5.7	0.43	HT
5 sites (R)	no tc			183.6	−40.9	103	7.6	2.35	HT
south	tc			163.0	−57.2	122	7.0	2.35	HT
11 sites (R)	no tc			175.5	−63.1	8	17.6	1.30	HT
all area	tc			155.4	−59.1	104	4.5	1.30	HT
Mean directions: Craco area									
4 sites (N)	no tc			347.0	56.6	23	19.5	1.11	LT
	tc			350.2	39.4	16	23.5	1.11	LT

TABLE 1 (continued)

Site	N	D	I	D(tc)	I(tc)	k	α_{95}	Int	Comp
Mean directions: Craco area (continued)									
2 sites (N)	no tc			270.5	71.2	22	56.7	1.09	HT
	tc			328.4	55.6	100	25.2	1.09	HT
4 sites (R)	no tc			164.0	−57.8	13	26.2	0.32	HT
	tc			161.8	−52.5	38	15.2	0.32	HT
6 sites (NR)	no tc			328.6	65.8	11	21.5	0.58	HT
	tc			337.5	53.7	48	9.8	0.58	HT

N = number of samples; D(tc), I(tc) = declination, inclination (including bedding tilt correction); k = precision parameter; $\alpha 95$ = semi-angle of cone of confidence on the 95% level; Int = mean intensity value calculated per site (in mA/m); Comp = component, where LT means low-temperature (100 − 300°C) and HT high-temperature (340–450°C) component.

nent which is removed at 100°C. A first NRM component is removed between 100 and 300–320°C, and is further referred to as a low-temperature (LT) component (Fig. 3C, D). A second component is removed between 300–320 and 430°C, defined here as high-temperature (HT) component (Fig. 3). This distinction in two NRM components is the same as observed for the clays overlying the Apulia–Gargano foreland (Scheepers, 1992).

The normal polarity LT component has a small overlap in blocking temperature spectrum (250–300°C) with the HT component (see, for example, Fig. 3C). Four sites from the area north of Sant'Arcangelo show such a normal polarity LT component (Table 1). The mean direction, calculated by giving equal weight to the site mean LT directions reveals the highest value for the precision parameter if no correction for bedding tilt is applied (Fig. 4A, B; Table 1): $D_m = 8.4°$ and $I_m = 39.9°$ ($k = 65$). The normal polarity LT component has therefore been acquired after bedding tilt and is probably caused by weathering. According to its direction it must be of subrecent (secondary) age, although the inclination is much shallower than the inclination (57.5°) of the geocentric axial dipole (GAD) field for the present latitude of the basin. The seven remaining sites in the Sant'Arcangelo basin show a reversed polarity LT component (Fig. 3B, D; Table 1); discrimination of this component is arbritary. The mean direction of these reversed polarity LT components shows the best concentration after bedding tilt correction has been applied (Fig. 4A, B; Table 1). Clustering is still not good after bedding tilt correction ($k = 21$), but better than before ($k = 9$; Table 1). Consequently, the reversed polarity LT component has been acquired before bedding tilt of the sediments and during an interval of reversed polarity, and is considered to be of "primary" origin.

In contrast to the LT component, the HT component shows a well-defined reversed polarity for all sites from the Sant'Arcangelo basin (Fig. 3; Table 1); there is no relation with the polarity of the LT component, for all sites the NRM decay from 320°C onwards is similar (Fig. 3). The HT directions cluster excellently after bedding tilt correction has been applied (Fig. 4C, D; Table 1); they show a positive fold test (McFadden, 1990) and maximum ($k = 104$) statistics at 100% unfolding: $D_m = 155.4°$ and $I_m = -59.1°$ for the eleven sites from the Sant'Arcangelo basin. The HT magnetisation has been acquired before tilting of the sediments and is considered to be of primary origin. Clustering of the HT component is much better than that of the reversed polarity LT component. The HT components of these sediments do not show an inclination error: the average inclination is 59.1° (Table 1; Fig. 4), which does not deviate significantly from the GAD inclination for the present latitude (57.5°). The origin of the HT magnetisation will be discussed later on the basis of the rock magnetic observations.

The reversed polarity and the inferred biostratigraphic age imply that these clays are lower Pleistocene in age (Matuyama Chron). It appears that there is a 25° counter-clockwise rotation relative to the magnetic North for the Sant'

Arcangelo clays; this value is similar to the results of Sagnotti (1992). It further appears that the mean HT direction for sites SAG 1–6 from the northern area gives a different rotation (31.5°) with respect to sites SAG 7–11 from the southern area (17.0°). The data of Sagnotti (1992) agree well with this observation; his five sites directly to the north of Sant'Arcangelo imply 28° of rotation: $D_m = 152.0°$ and $I_m = -59.8°$ ($k = 203$); his four sites directly south of Sant'Arcangelo imply 18° of rotation: $D_m = 161.7°$ and $I_m = -57.3°$ ($k = 109$)

For the northern area, comparing the HT di-

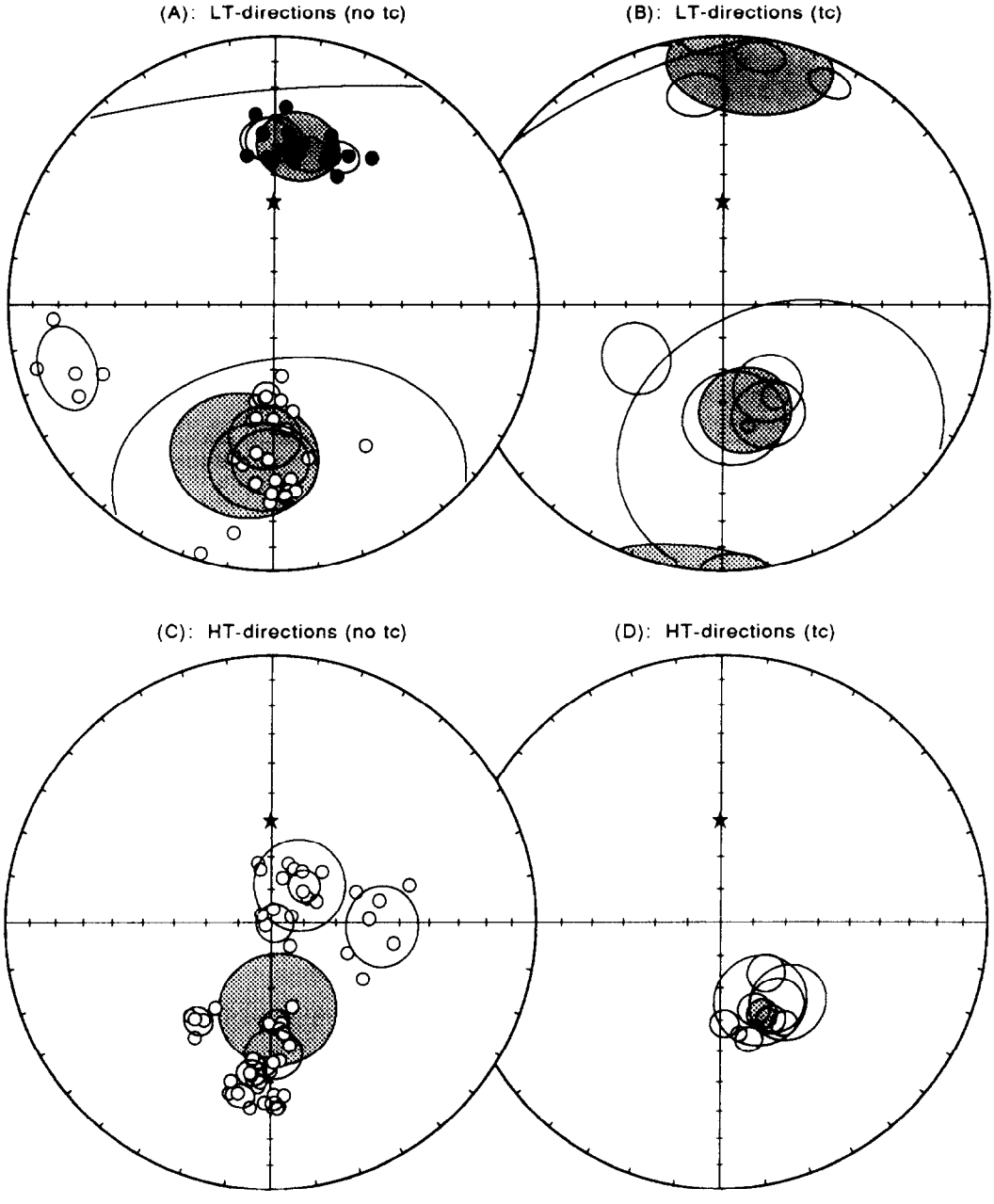


Fig. 4. Equal area projection of different NRM components of samples from the Sant'Arcangelo basin. In the left-hand diagrams the circular symbols represent the individual sample directions; • denotes downward projections, ○ denotes upward projections. The circles give Fisher's (1953) α_{95} for the different site means. The shaded area gives the α_{95} of the overall mean direction using the site means. ★ denotes the geocentric axial dipole field direction for the sampling area.

rection of site SAG 3—which has a primary LT magnetisation—with the HT directions of sites SAG 1, 4, 5 and 6 (having a secondary LT magnetisation), shows that the direction of the HT magnetisation is hardly or not influenced by the LT magnetisation (Table 1). The mean direction for site SAG 3 is: $D_m = 147.6°$ and $I_m = -53.1°$ ($k = 398$). The mean direction for the four sites SAG 1, 4, 5 and 6 is: $D_m = 147.6°$ and $I_m = -63.1°$ ($k = 151$). Site SAG 2 was not used here because its NRM characteristics are completely different from the other ten sites. From the demagnetisation diagrams (Fig. 3) it can be seen that there is no overlap in blocking temperature spectrum of the two components above 340°C. However, the intensity of the HT magnetisation is much lower for sites with a secondary LT component (0.33 mA/m) than for the sites with a primary (here: reversed) LT component (1.86 mA/m). Apparently, weathering has removed part of the HT magnetisation, but did not affect its direction.

The Craco area

At the most external part of the Apenninic belt system, just west of the Bradano Unit, we sampled an area south of the abandoned village of Craco (Fig. 2). Six sites with different bedding tilts were sampled here in the clayey Gravina

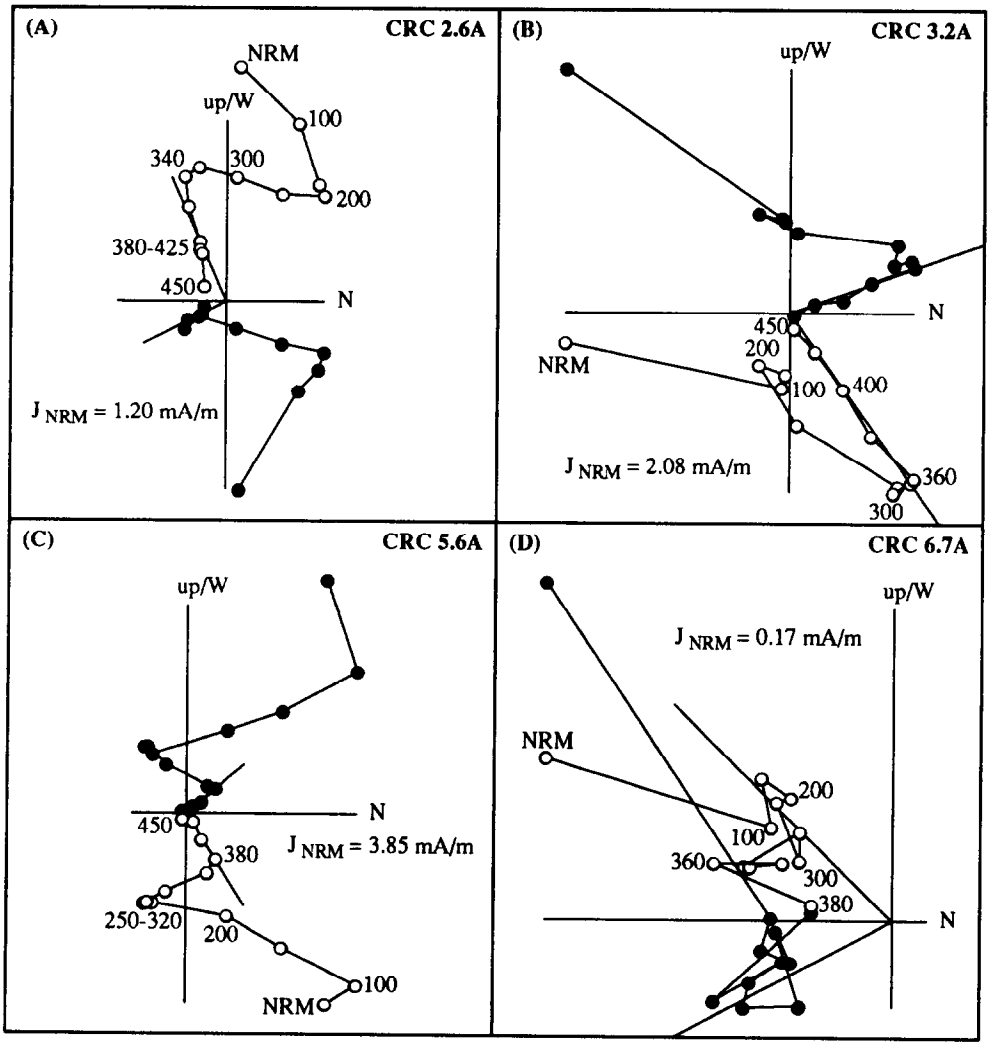


Fig. 5. Stepwise thermal demagnetisation diagrams of samples from the Craco area. All diagrams include bedding tilt correction. Symbols are the same as used in Fig. 3.

formation (foglio 200, Carta geologica d'Italia). Sites CRC 1–3 and 5 have a tilt between 22 and 39° to the north (northeast), while CRC 4 and 6 have one of 21–25° to the southeast (Table A2). All sites are in blue-coloured clays; sites CRC 1, 4 and 6 showed grey-brown coloured spots indicating a slight degree of weathering.

There are two types of demagnetisation behaviour for these six sites: there are four sites with complex multi-component behaviour (CRC 1–3 and 5) and there are two sites with low intensities showing scattered one-component behaviour (sites CRC 4 and 6). For the low-intensity sites, only a reversed polarity magnetisation is removed from 200 up to 400°C (Fig. 5D); the scatter is mainly caused by the relatively weak remanences (for example $J_{HT} = 50$ mA/m for site CRC 6). The other four sites not only have higher initial intensities, but also show clearly more NRM components; the outcrops of these sites are the fresh (blue coloured) clays. The NRM components from these fresh sites are: a LT component removed between 100 and 200°C; another component removed between 200 and 340°C (medium temperature or MT component); and finally a HT component between 340–360 and 450°C (see for example Fig. 5A, C). The HT component is present in all sites and gives a positive fold test (see Fig. 6; Table 1), increasing the precision parameter ($k = 11$ to $k = 48$). The mean direction of the HT component for the Craco area implies a counter-clockwise rotation of 22.5°; this value is quite similar to the rotation of the Sant'Arcangelo basin (24.5°, see above; Sagnotti, 1992).

The other NRM components present in the fresh outcrops of these clays do not affect the HT direction. Some sites have a medium-temperature (MT) component with opposite polarity relative to the HT component; site CRC 2 has a reversed polarity HT component and a normal polarity MT component and a reversed LT component (Fig. 5A). Further, site CRC 5 has a normal polarity HT component; the MT component of this site is southerly directed (Fig. 5C), but because of the major interference with the LT and HT component (both normal polarity) the inclination of this MT component is variable. This MT component is removed between 200 and 340–380°C (Fig. 5B, C), and shows in this temper-

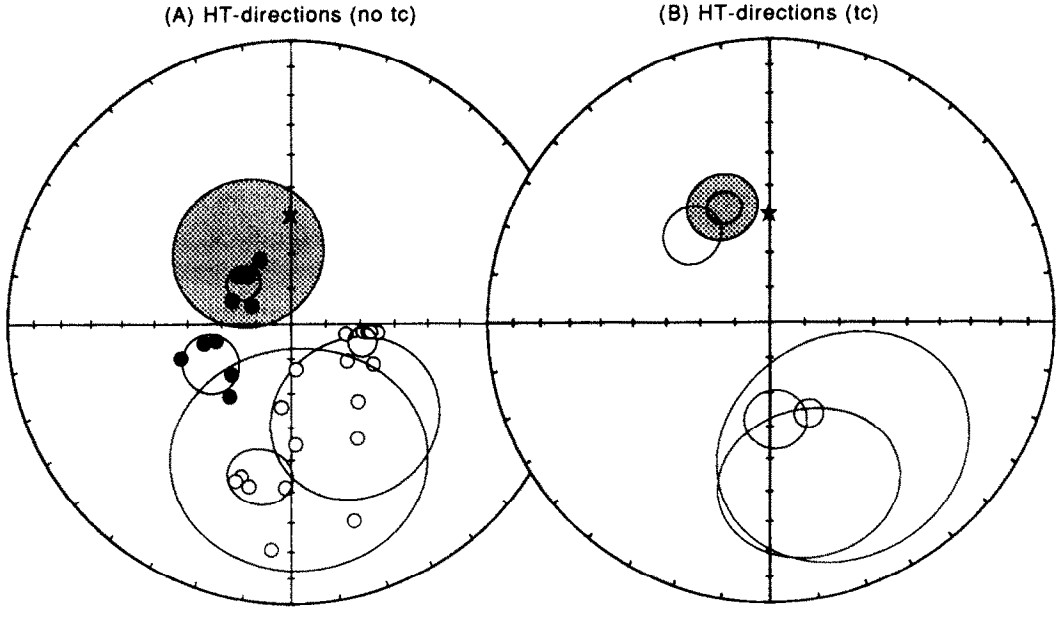


Fig. 6. Equal area projections of HT magnetisation components of the Craco area (CRC). (A) In-situ directions (all samples and site mean directions including α_{95}). (B) Tilt corrected directions (site mean directions including α_{95}). Reversed LT site mean directions have been inverted to compute the overall mean direction (shaded ellips). See also caption to Fig. 4.

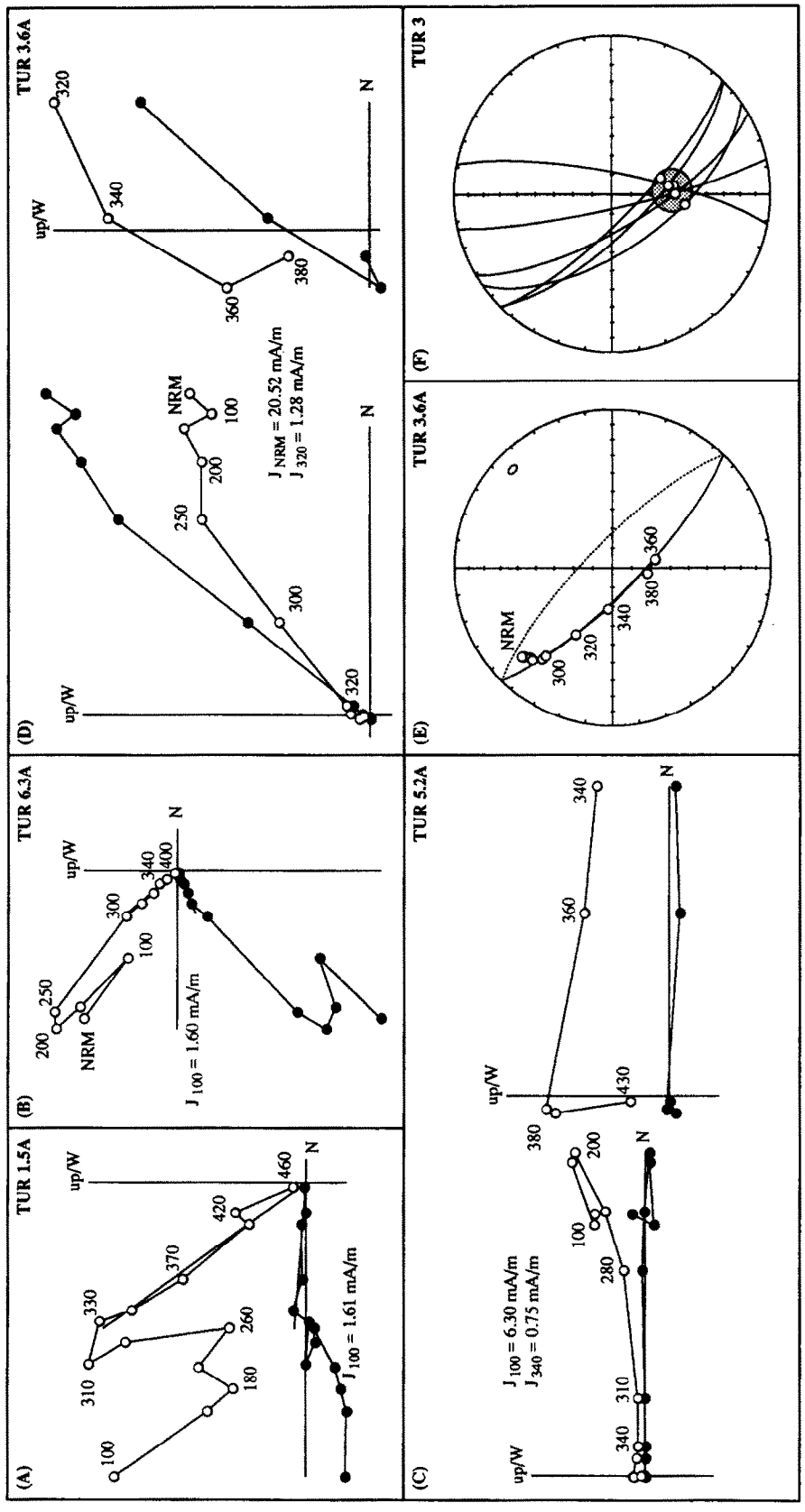


Fig. 7. Stepwise thermal demagnetisation diagrams of samples from the Tursi area (TUR) including bedding tilt correction. The equal area projection (E) shows the great circle of sample TUR 3.6A (D); the equal area projection (F) shows the great circle planes of the samples belonging to site TUR 3, with the calculation of the "hidden" final component. Other symbols are the same as in Figure 3.

TABLE 2

Site mean directions and overall mean directions from the sites of the Bradano Unit sampled in the Matera area

Site	N	D	I	D(tc)	I(tc)	k	α_{95}	Int	Comp
Tursi section, direct approach									
TUR 1	5	13.3	72.9	21.8	67.5	48	11.2	1.04	LT
TUR 2	6	22.5	64.1	26.9	58.4	29	12.6	0.99	LT
TUR 3 *	6	330.6	−20.2	328.3	−21.5	5	34.8	35.24	LT
TUR 4 *	6	324.4	29.6	327.8	28.7	6	29.4	7.35	LT
TUR 5 *	4	345.5	2.2	345.6	−0.7	3	64.6	5.68	LT
TUR 6	6	328.8	53.4	342.5	54.5	8	24.8	14.63	LT
TUR 7	5	342.4	55.8	335.2	51.9	14	21.1	14.04	LT
TUR 1	6	171.8	−58.9	179.3	−55.1	439	3.2	1.08	HT
TUR 2	6	175.2	−56.8	181.8	−52.7	295	3.9	0.90	HT
TUR 3–5	5	189.4	−74.3	199.5	−69.0	12	23.2	0.39	HT
TUR 6	6	129.7	−43.3	138.8	−48.0	18	16.1	0.30	HT
TUR 7	5	189.0	−43.8	183.0	−42.8	76	8.8	0.44	HT
Tursi section, remagnetisation circles									
TUR 1	6	175.5	−65.5	184.6	−61.3		5.4		HT
TUR 2	6	175.2	−58.5	182.2	−54.4		4.0		HT
TUR 3	6	167.6	−62.1	176.5	−58.5		10.6		HT
TUR 4	4	189.9	−53.2	194.3	−48.1		12.4		HT
TUR 5	4	160.2	−70.1	173.9	−67.0		5.9		HT
TUR 6	6	160.4	−51.9	173.0	−51.1		11.0		HT
TUR 7	6	177.2	−56.4	168.5	−53.9		4.4		HT
Tursi section, mean directions									
Direct approach:	4		no tc	352.0	63.3	30	17.0	7.68	LT
			tc	358.4	60.1	28	17.6	7.68	LT
Direct approach:	5		no tc	168.2	−57.6	18	18.6	0.62	HT
			tc	174.4	−55.2	25	15.6	0.62	HT
Remag. circles:	7		no tc	173.0	−60.0	90	6.4		HT
			tc	179.4	−56.6	98	6.1		HT
Pisticci section, direct approach									
PST 2	6	5.1	53.9	10.4	48.3	163	5.3	0.74	LT
PST 8	6	359.5	42.7	3.6	37.6	360	3.5	0.78	LT
PST 9	6	357.1	47.7	2.1	42.8	103	6.6	1.19	LT
PST 1	5	3.6	52.2	8.7	46.8	90	8.1	1.92	HT
PST 3	6	156.4	−52.8	164.3	−49.8	64	8.4	1.38	HT
PST 4	5	1.7	60.4	8.8	55.0	281	4.6	2.54	HT
PST 9 *	6	121.4	−69.3	140.3	−69.8	21	15.0	0.10	HT
PST 13	3	5.5	55.2	11.0	49.6	90	13.0	0.64	HT
PST 14	3	11.9	59.1	17.2	53.1	44	18.9	0.22	HT
Pisticci section, remagnetisation circles									
PST 8	6	176.9	−48.6	182.1	−43.7		4.0		HT
PST 9	6	166.1	−64.1	176.7	−59.9		6.0		HT
Pomárico area, direct approach									
POM 1	6	348.5	58.6	348.5	58.6	105	6.6	0.58	HT
POM 2	6	359.6	62.5	359.6	62.5	136	5.8	1.46	HT
POM 3	6	356.0	53.7	349.8	59.9	319	3.8	1.24	HT
Mean directions									
Pisticci	3	0.3	48.1	5.2	43.0	171	9.5	0.90	LT
Pisticci (N)	4	5.6	56.8	11.4	51.2	338	5.0	0.74	HT

TABLE 2 (continued)

Site	N	D	I	D(tc)	I(tc)	k	α_{95}	Int	Comp
Mean directions (continued)									
Pisticci (R)	3	166.9	−55.5	174.5	−51.4	65	15.5	1.33	HT
Pisticci (NR)	7	357.5	56.6	4.2	51.6	88	6.5	0.85	HT
Pomárico (N)	3		no tc	354.7	58.3	236	8.0	1.09	HT
			tc	352.4	60.4	519	2.4	1.09	HT

* Denotes that this site was not used for calculation of mean direction. Other symbols the same as in Table 1.

ature interval a more or less linear decay (CRC 2 and 5). Because the sites which show this behaviour were the most fresh, it is possible that this component is carried by iron sulphides (probably pyrrhotite) disturbing the NRM up to 340°C. Because of the major interference with the other (non-sulphide) components, it is not possible to determine this component with sufficient accuracy.

The Matera area, the Bradano Unit

The Tursi area

The Tursi area is the most southerly of the three sampled areas belonging to the Bradano Unit in the Matera area (Fig. 2). We sampled seven sites (TUR 1–7) in the Tursi section. Sites TUR 1–2 were sampled in a fresh outcrop in the silty lower part of the marine clay sequence. The sites TUR 3–7 are situated along the road to the east (to locality San Maria d'Anglona) in the blue-coloured clayey middle part of the formation. The relative stratigraphic position of the sites was determined; the age of the sampled interval is between 0.88 and 0.78 Ma (see Appendix). Sediments of the Tursi area show a moderate (7°) and rather uniform tilt to the north-northeast (Table A2).

The NRM characteristics of the sediments in the Tursi section differ slightly but significantly from the previous areas: the behaviour is less uniform, although maximum blocking temperatures are similar (430°C). For sites TUR 1–2 three distinct NRM components can be distinguished (Fig. 7A). Firstly, a component with reversed polarity was removed between 100 and 200–250°C; this component is not present in all samples (see Scheepers and Zijderveld, 1992). Secondly, a low-temperature (LT) normal polarity component is removed between 200–250 and 300–320°C. Finally, a reversed polarity HT component was removed between 300–320 and 430°C (see also Fig. 7A). The HT directions show a better concentration than the other components (Table 2), with a precision parameter of 300–400. It is evident that the mean direction of this HT component does not reveal any rotation.

For sites TUR 3–5, a very strong and nearly randomly oriented LT component is present (Fig. 7B, D; Table 2). A reliable determination of the HT component for these sites is hardly possible (Fig. 7D), since it is largely overprinted by the LT component, which is stronger by one or two orders of magnitude. Still, it can be seen that the HT component has a reversed polarity (Fig. 7D; Table 2). Since a more or less reliable direction could only be determined from a few specimens of each site, TUR 3–5 was taken as one site to calculate the HT mean direction, but it is still badly determined ($k = 12$). The great circles through the magnetisation directions are very well determined for these sites (Fig. 7E); between 250 and 360°C great circles have maximum angular deviations between 1 and 10°. These great circles were calculated for all samples demagnetised per site; the resulting great circles converge excellently (Fig. 7F). We have used the great circle analysis method of McFadden and McElhinny (1988). We imposed no sector constraints because the planes make fairly large angles with each other; we only impose a reversed polarity as an approximate direction. This procedure was also performed for sites TUR 1–2; the great circles, calculated for the 250–360°C interval converge here also very well. Because we have very reliable HT directions here (Table 2), we can use the results from the two different approaches as a

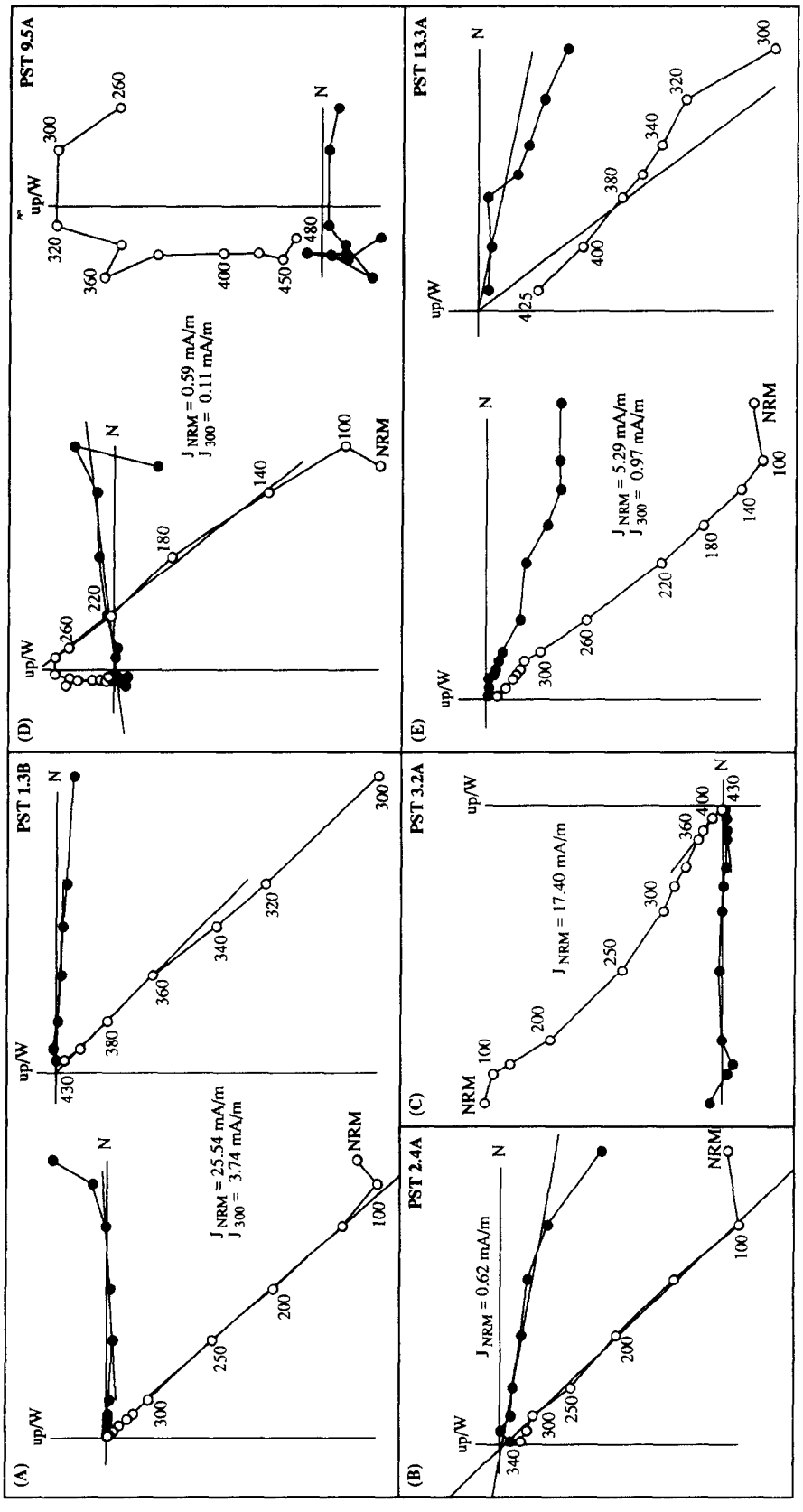


Fig. 8. Stepwise thermal demagnetisation diagrams of samples from the Pisticci area (PST). See also caption to Fig. 3.

check for the great circle method. The directions of the two methods are very similar (Table 2). For sites TUR 3–5 the great-circle-derived HT directions are in accordance with the results from sites TUR 1–2 (Table 2).

The upper site of the Tursi section (TUR 6) gives very peculiar HT directions; the mean direction of the HT components is more than 40° to the west (Table 2). Because the precision parameter is low ($k = 18$), there is some doubt on the stability of the component here. The great circles converge at a complete different direction (Table 2), which is in accordance to the results of the other sites. The other site in the upper part of the section (TUR 7) has a HT direction which agrees with the directions of the lower part of the section. We further use the HT components calculated from the great circle method for the Tursi section; because of the major interference with other NRM components, the HT component has a less reliable uncertain direction with the direct approach. The mean direction for the seven sites in the Tursi section is: $D_m = 179.4°$ and $I_m = -56.6°$ ($k = 98$, Table 2), implying no rotation for the Tursi section.

The Pisticci section

The Pisticci section is also in the Bradano Unit, to the north of the Tursi section (Fig. 2). In the main section eight sites were collected; further three sites were sampled down in the Cavone valley. From the main section, sites PST 1–4 and 8–9 are from the middle part of the exposed Pleistocene clays (Argille Subappennine—foglio 201, Carta geologica d'Italia), sampled 2 km south-southeast of the village of Pisticci. Sites PST 13–14 are from the top part of the section, just east of the village of Pisticci. Down in the Cavone valley (Fig. 2), three sites (PST 5–7) were collected in the lowest part of the section. Bedding orientation could not be established here; this part of the section is tectonically very disturbed; we use the paleomagnetic results only for the age estimate of the section. Most sites in the Pisticci section were drilled in blue-coloured clays except sites PST 2, 6, 8–9 and 14. We drilled site PST 2 in clays with a brown/red colour indicating extensive weathering, with the purpose to compare the results with the fresh sites. At site PST 1, synsedimentary slumps are present. The bedding tilt for the Pisticci section is approximately 7° to the north-northeast (Table A1). The sites from the middle part of the section can be dated around 0.90 Ma (see Appendix).

The demagnetisation diagrams (Fig. 8) show the same characteristics as those of the Sant'Arcangelo basin (see above) and the Apulia–Gargano foreland (Scheepers, 1992). Apart from a viscous component, a LT component is removed between 100 and 300–320°C, and a HT component is removed between 300–320 and 430°C. The distinction between these components is only evident from the sites which show increased weathering (e.g., PST 9.5A, Fig. 8D). Site PST 9 has a normal polarity LT component and a very low intensity reversed polarity HT component (Fig. 8D; Table 2); the direction of the HT component is badly determined and is clearly affected by weathering to a larger extent than was found for the sites in the Sant'Arcangelo area. We used the great circles to calculate the HT direction here (Table 2); also for site PST 8 we used the great circles to determine the "hidden" HT direction.

The site taken from the weathered outcrop (PST 2) has lost its magnetisation almost completely at 300–340°C (Fig. 8B); thus only the (normal) LT component is present here. The (in-situ) direction of this component has a present-day field direction (Table 2) and apparently extensive weathering has completely removed the HT magnetisation. This process was already suggested from the observation in the Sant'Arcangelo basin that increased weathering removed part of the HT magnetisation. Evidently, there are various stadia in the process of weathering and its influence on the different NRM components.

The fresh sites of the Pisticci area are similar to the fresh sites from the Sant'Arcangelo basin: intensities are of the same order of magnitude and the samples have similar maximum blocking temperatures. In the Pisticci section both normal and reversed polarities are found for the HT component. Site PST 3 (Fig. 8C) has a reversed polarity. The others, PST 1 and 4 (Fig. 8A) show a normal polarity (Table 2). The HT directions

are clearly not antipodal here. The primary origin of both polarities is supported by the observation that both normal and reversed polarity magnetisations have the same characteristics in NRM decay as well as high intensities (Fig. 8A, C; Table 2, compare Table 1). Therefore, the normal polarity sites (PST 1 and 4) are considered to represent a true normal polarity event, although we notice that the (in-situ) direction is close to the present-day field direction (Table 2). According to biostratigraphic age estimates this normal polarity event represents the Jaramillo subchron. Site PST 1 shows consistent directions of remanence, although slumps are present. The NRM is not disturbed by the slumping and indicates that either (synsedimentary) "resetting" of the remanence occurred due to the probably high water content involved with the slumping (see also Langereis and Hilgen, 1991), or that there is a time-lag in recording the geomagnetic field, e.g., due to authigenic formation of the remanence carrying minerals (Van Hoof and Langereis, 1991).

The sites from the upper part of the section, PST 13–14, have a clear normal polarity HT magnetisation (Fig. 8E). The direction is not significantly different from the normal polarity sites from the middle part of the section (Table 2). The mean direction for the four normal polarity sites from the Pisticci section and the mean (inverted) direction for the three reversed polarity sites make an angle of 10.5°, the reversal test (McFadden and McElhinny, 1990) is positive, classification "C".

The Pomárico area

North of the Pisticci area and in the Bradano Unit, between the Basento and Brádano rivers (Fig. 2), we collected three sites (POM 1–3) in

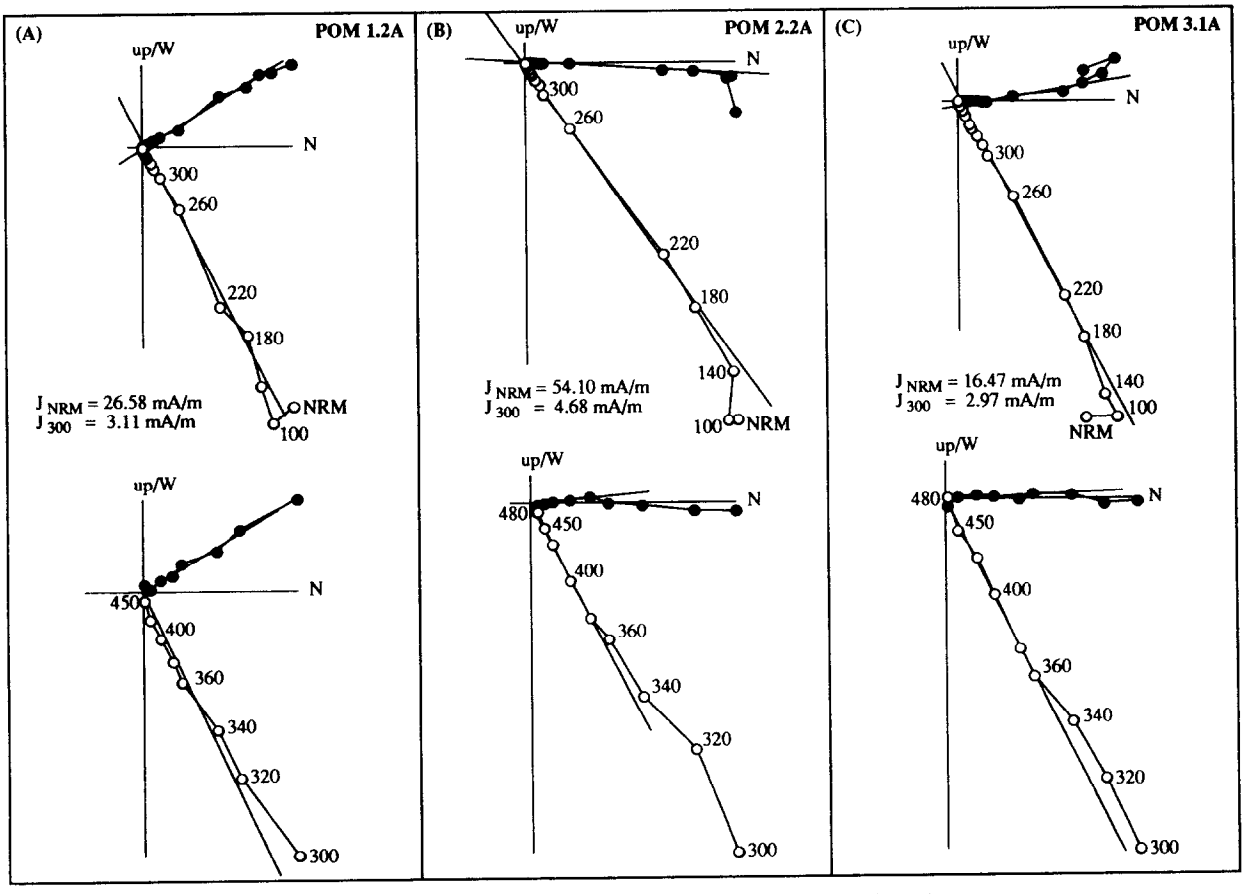


Fig. 9. Stepwise thermal demagnetisation diagrams of samples from the Pomárico area (POM). See also caption to Fig. 3.

the same Pleistocene clays (Argille Subappennine —foglio 201, Carta geologica d'Italia). The bedding is sub-horizontal for sites POM 1–2, site POM 3 has a bedding tilt of 7° to the south-southeast (Table A2). All sites were sampled in very fresh clays.

The NRM characteristics are the same as for the fresh sites from the other areas. All three sites have a normal polarity HT component (Fig. 9), consistently showing a very small westerly rotation of 7.5° (Table 2). The primary origin is supported by the consistent NRM characteristics.

Comparing HT directions from the Matera area

The three areas from the western part of the Matera area and belonging to the Apenninic belt show a clear counter-clockwise rotation as derived from their HT site mean directions (Fig. 10; Table 3). The seventeen sites have a mean direction implying an average rotation of 24°, which is exactly the same if the mean direction of the three areas is calculated (Table 3). The normal and reversed polarity sites were used to perform a reversal test; the mean direction for the two normal polarity sites and the (inverted) mean direction for the fifteen reversed polarity sites have an angle of 5.2°, the reversal test (McFadden and McElhinny, 1990) is positive, classification "C".

The three areas from the eastern part of the Matera area and belonging to the Bradano Unit have in total seven normal polarity sites. The clustering for the sites is not as good as for the Apenninic belt; the seventeen sites have a mean direction, similar to the mean direction of the three areas (Table 3). The HT mean directions do not significantly deviate from north–south (Fig 10; Table 3). The mean declination (D_m) for the seven normal polarity sites is 4.4°, and for the seven reversed polarity sites 177.8°; mean inclination is nearly the same (55°; Table 3). The angle between normal and reversed polarity is 3.7°, the reversal test is positive, classification "B". These directions are not significantly different from the earlier observed data from the Apulian Unit

TABLE 3

Summary of main results from this paleomagnetic study in the Matera area in southern Italy

N	*D*(tc)	*I*(tc)	*k*(tc)	Pol.	Locality	Age estimate
Apenninic belt (this study)						
6 sites	328.5	60.3	140	R	Sant'Arcangelo (SAG) north	2.02–0.70 Ma
5 sites	343.0	57.2	122	R	Sant'Arcangelo (SAG) south	2.02–0.70 Ma
6 sites	337.5	53.7	48	NR	Craco (CRC)	2.02–1.45 Ma
17 sites	336.2	57.2	73	NR	Apenninic belt sites	
2 sites	328.4	55.6	100	N	normal polarity sites	
15 sites	337.3	57.4	70	R	reversed polarity sites	
3 areas	336.6	57.2	254	NR	Apenninic belt localities	
Bradano Unit (this study)						
7 sites	359.4	56.6	98	R	Tursi section	0.89–0.82 Ma
7 sites	4.2	51.6	88	NR	Pisticci section	1.05–0.90 Ma
3 sites	352.4	58.3	519	N	Pomárico area	< 0.99 Ma
17 sites	0.5	55.3	91	NR	Bradano Unit sites	
7 sites	4.4	55.5	94	N	normal polarity sites	
10 sites	357.8	55.1	87	R	reversed polarity sites	
3 areas	359.1	56.3	218	NR	Bradano Unit localities	
Apulian Unit (from Scheepers, 1992)						
8 quarries	358.2	56.5	215	N	Apulian Unit sites	early Pleistocene
2 quarries	0.2	59.1	2162	N	normal polarity sites	
6 quarries	357.6	55.6	167	R	reversed polarity sites	
4 areas	359.4	57.0	453	NR	Apulian Unit localities	

N = number of sites/samples; *D*(tc), *I*(tc) = declination, inclination (after tectonic correction); *k*(tc) = precision parameter (after tectonic correction); *k*(tc)/*k* = gives the improvement of clustering after tectonic correction; Pol = polarity.

(Scheepers, 1992). Inclination values are constant 55–57° for these mostly lower Pleistocene clays and hence there is no deviation from the geocentic axial dipole field for the present latitude of the area.

Some rock magnetic observations

General observations

In order to support the paleomagnetic interpretation of the studied sediments and to obtain additional information on the origin of the NRM, some rock magnetic experiments were performed. Several sites of the Sant'Arcangelo basin and the Pisticci section were used to study the influence of weathering on the rock magnetic properties. The NRM of these late Tertiary and Quaternary sediments is removed between 100 and 430–450°C. NRM decay curves (Fig. 11) were determined for six different sites, by taking three examples from very fresh sites (SAG 3, 11 and PST 1), two examples from slightly weathered sites showing a secondary LT component and a reliable HT direction (SAG 4 and 6) and one example from a site entirely weathered without any HT component remaining (PST 2). The NRM decay curves were determined after stacking all available demagnetisation diagrams of this site (method explained in Scheepers and Zijderveld, 1992). The decay curves demonstrate that the fresh sites lose most of their magnetisation between 150 and 250°C (Fig. 11). The absolute and the relative amount of NRM decay in this temperature interval is slightly more in the fresh sites. The decay of magnetisation is decreasing from 250°C onwards. There is no peak in NRM decay which may be linked to the blocking temperature of an additional magnetic mineral.

For the sites from the Sant'Arcangelo basin, the behaviour of the low-field susceptibility X_{LF} during thermal demagnetisation was monitored. The X_{LF} decreases strongly between 150 and 320°C (Fig. 12) for samples from the fresh sites (SAG 3, 10 and 11, with a primary LT component) from values between $(500–700) \times 10^{-6}$ and $(300–350) \times 10^{-6}$. These sites also showed a major drop in NRM intensity for this trajectory (Fig. 11); their mean initial NRM intensity is 9.4

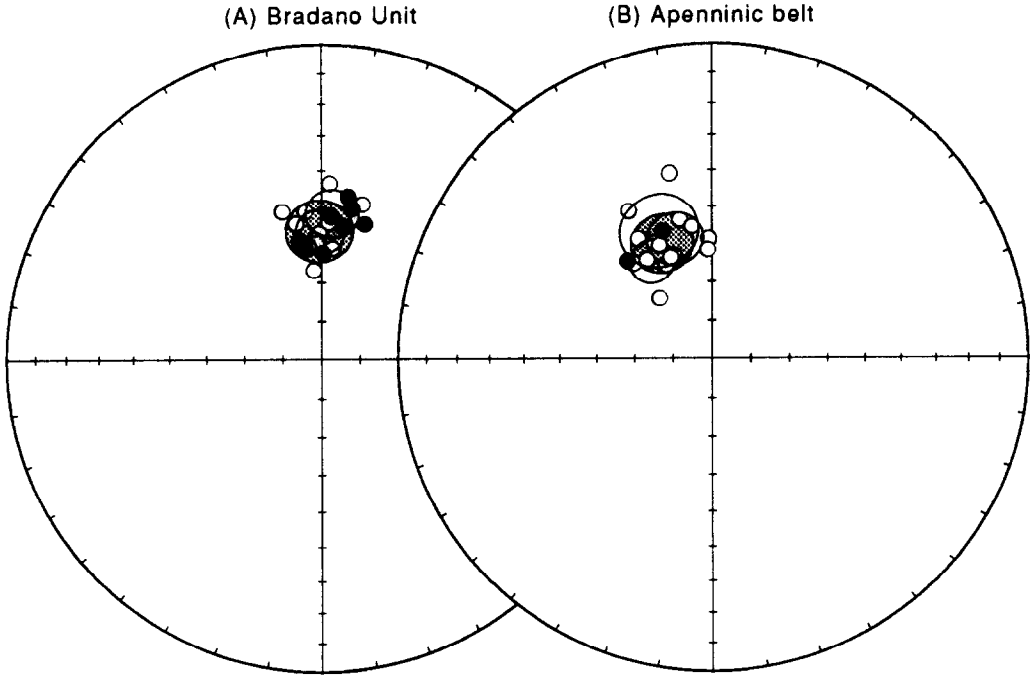


Fig. 10. Site mean directions for HT magnetisation for sites from the Bradano Unit (A) and the Apenninic belt (B) after inverting reversed polarity directions (circles). The circles give the α_{95} for the different areas. The shaded ellips gives the α_{95} of the overall mean direction using the area means.

mA/m. This decrease in X_{LF} between 150 and 320°C was observed for all sites in the Matera area with NRM intensities above 5 mA/m and/or initial X_{LF} values above 300×10^{-6} (SI-units).

The samples from the slightly weathered sites (SAG 4–6) show almost no or a slight decrease in X_{LF} during thermal demagnetisation. The mean initial NRM intensity for these sites is appreciably lower, only 3.3 mA/m. Apparently oxidation due to subrecent weathering of the remanence carrying minerals unblocking at ±300°C has removed the primary LT component. This same oxidation process was accelerated in the furnace during thermal demagnetisation; we observe therefore a decrease in X_{LF} and the intensity during thermal demagnetisation for the samples from the fresh sites. In the outcrops showing some weathering, the Earth's magnetic field introduced the secondary (normal) LT component during this oxidation process.

Thermomagnetic measurements

Thermomagnetic measurements are useful to distinguish between chemical alterations during thermal demagnetisation and true Curie temperatures for the unblocking interval. Such measurements were performed on samples from sites

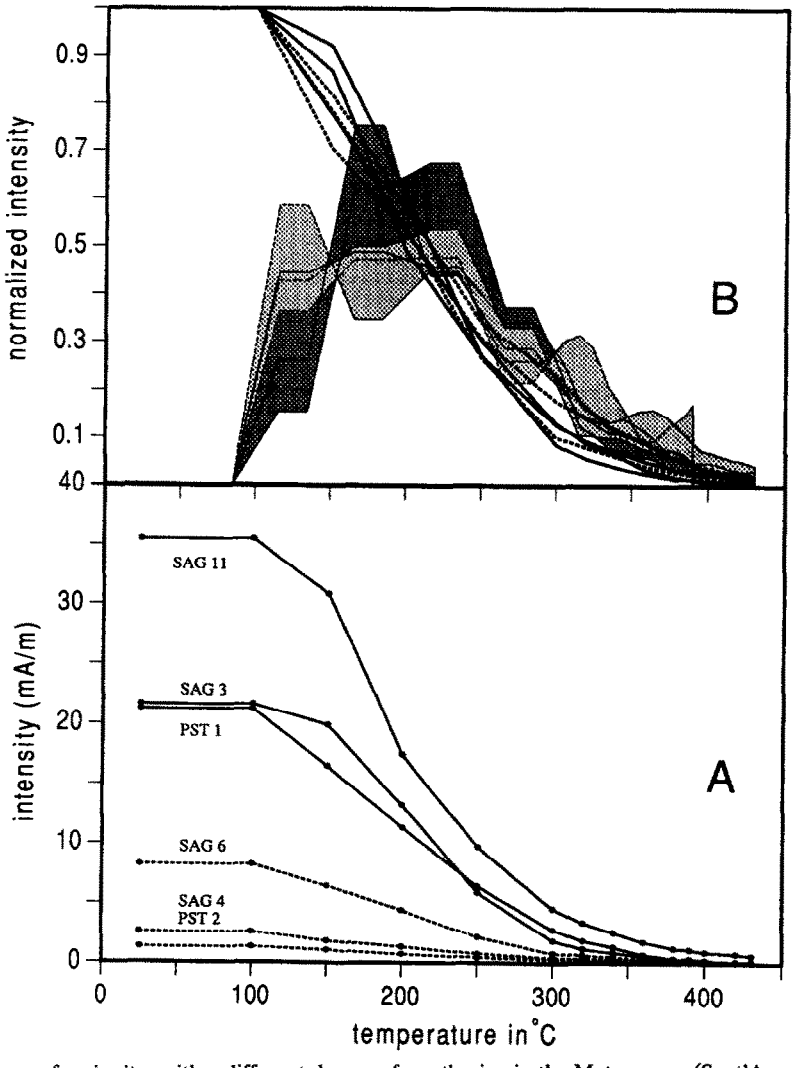


Fig. 11. NRM decay curves for six sites with a different degree of weathering in the Matera area (Sant'Arcangelo and Pisticci). (A) Decay curves. (B) Normalized decay curves and decay rate for normalized curves. Dark-shaded area denotes fresh sites; light-shaded area denotes weathered sites.

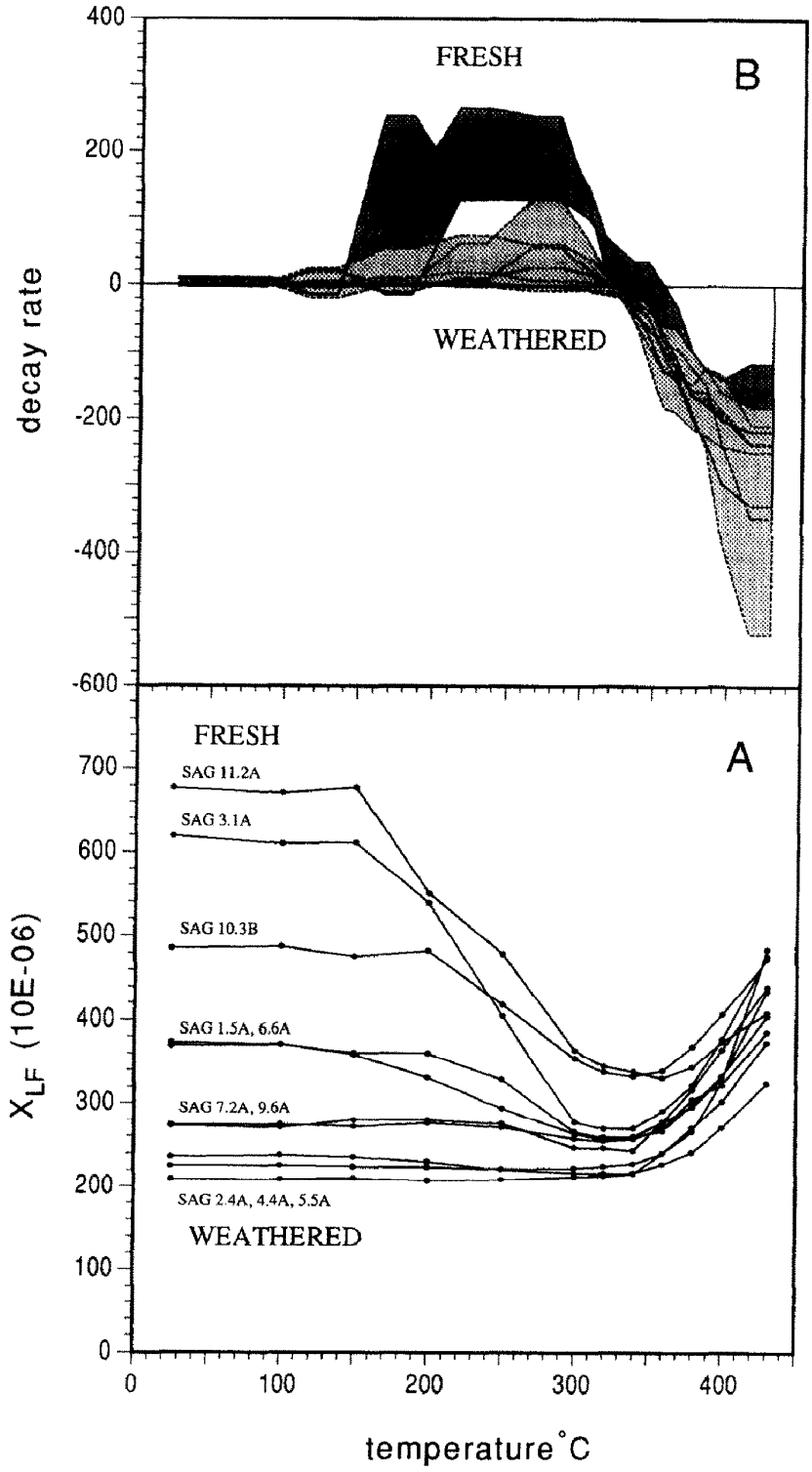


Fig. 12. (A) Initial (bulk) susceptibility versus demagnetisation temperature for samples of the Sant'Arcangelo basin. Upper part of (A) shows that the most fresh sites (*SAG 11*, *3* and *10*) have the largest initial susceptibility. A considerable drop in susceptibility occurs between 150 and 300°C. The sites containing a secondary LT component (*SAG 4*, *5*, *1* and *2*) have only a very small (or no) drop in susceptibility. An increase in susceptibility at temperatures above 340°C is observed for all sites. (B) gives the decay rate of the susceptibility. Dark-shaded area denotes fresh sites; light-shaded area denotes the weathered sites.

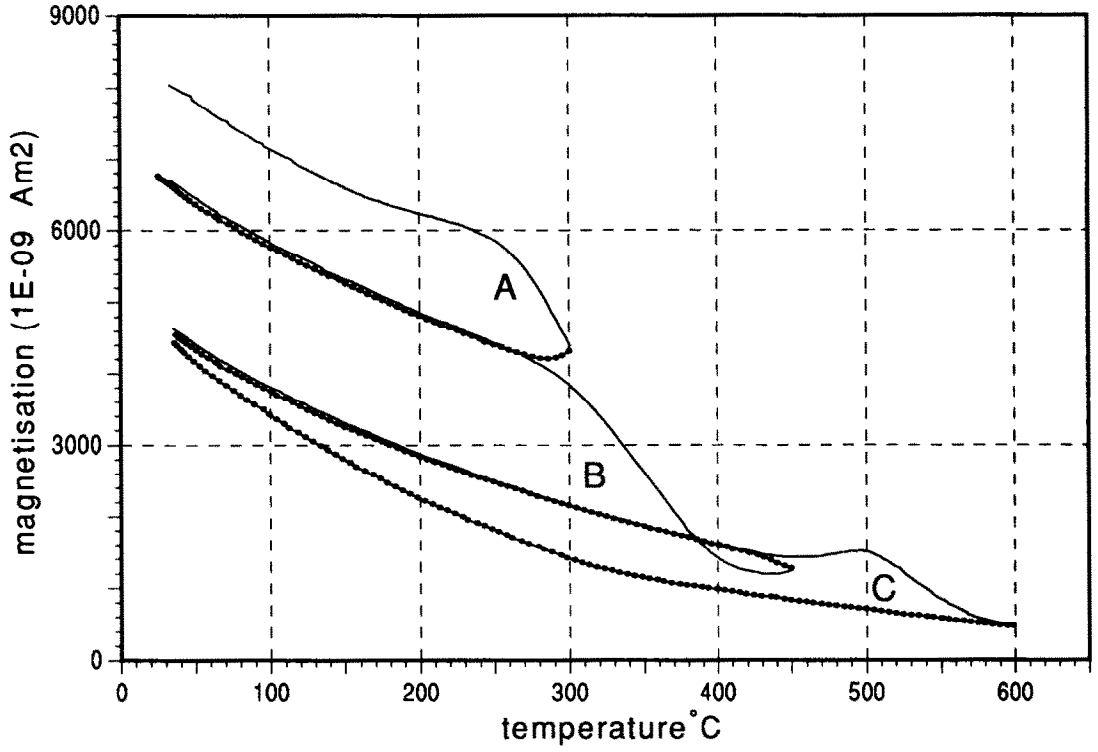


Fig. 13. Thermomagnetic curve measured in a cycling field (200–400 mT) for an unconcentrated clay sample (90 mg) from the Sant'Arcangelo basin (SAG 11.1B); three heating-cooling runs were done (*A*, *B*, *C*) up to increasing maximum temperatures (300, 450 and 600°C). Solid line gives the heating curves (rate 6°C/min), dotted line gives the cooling curves (rate 10°C/min). There is a major decrease in magnetisation between 250 and 430°C.

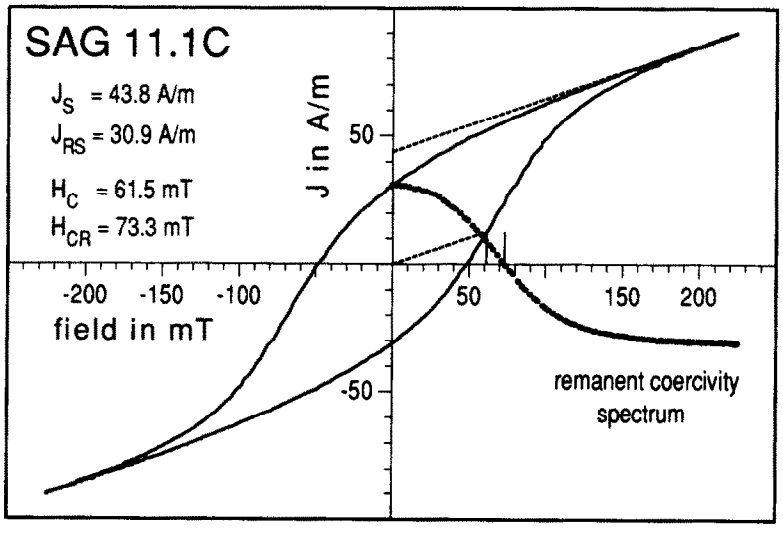


Fig. 14. Magnetic hysteresis loop for clay sample SAG 11.1C. Dotted line gives the remanent coercivity spectrum of the sample. The linear relation of magnetisation to applied field (at higher fields) is caused by the paramagnetic contribution of the clay minerals.

SAG 11, 4 and PST 2, representing the different stadia of weathering. A modified horizontal translation type Curie balance, employing a cycling magnetic field (see Mullender et al., 1993) was used. This novel technique considerably enhances the sensitivity and accuracy of thermomagnetic analyses, so that no concentration of magnetic minerals is necessary. The sample from

TABLE 4

Rock magnetic properties deduced from hysteresis measurements of clay samples from the Matera area

Specimen	H_{cr}	H_c	J_{rs}	J_s	(A)	(B)	X_{LF}	X_{HF}	d	(C)
$J_{rs} < 1$ A/m										
CRC 4.6B	78.6	44.1	0.47	0.62	1.78	0.76	206	169	37	79
CRC 6.1B	63.9	51.6	0.43	1.21	1.24	0.36	195	162	33	77
PST 2.4B	79.1	39.3	0.55	1.11	2.01	0.50	201	159	42	76
PST 9.2B	82.6	52.3	0.70	1.26	1.58	0.56	197	120	77	110
PST 14.1B	85.5	44.2	0.72	1.51	1.93	0.48	168	145	23	32
SAG 2.2B	131.6	47.9	0.53	1.46	2.75	0.36	225	182	43	81
$J_{rs} < 5$ A/m										
CRC 1.1B	74.3	59.3	3.92	5.72	1.25	0.69	242	169	73	19
CRC 2.5B	80.7	49.2	1.94	3.53	1.64	0.55	214	151	63	33
PST 13.4B	60.3	45.7	4.24	6.90	1.32	0.62	223	160	63	15
SAG 4.5B	71.4	40.0	1.28	3.33	1.78	0.38	244	178	66	52
SAG 5.5B	80.6	63.8	2.03	2.94	1.26	0.69	239	164	75	37
SAG 7.4B	84.5	70.5	3.47	4.71	1.20	0.74	298	177	121	35
SAG 8.6B	94.2	61.5	1.12	1.98	1.53	0.56	252	163	89	79
TUR 6.3B	72.5	61.2	3.46	4.33	1.19	0.80	260	185	75	22
$J_{rs} > 5$ A/m										
CRC 3.6B	67.7	56.2	34.1	49.2	1.20	0.69	493	219	274	8
CRC 5.1B	73.1	62.1	13.6	19.3	1.18	0.71	328	161	167	12
POM 1.1C	66.1	55.5	10.9	15.8	1.19	0.69	293	178	115	11
POM 2.6B	60.4	49.2	35.5	52.3	1.23	0.68	579	209	370	10
POM 3.4B	62.2	50.1	16.9	25.3	1.24	0.67	351	187	164	10
PST 1.3A	68.9	56.6	23.6	33.9	1.22	0.70	405	225	180	8
PST 3.2B	72.9	58.3	9.8	14.3	1.25	0.69	328	186	142	15
PST 4.1B	69.1	58.0	20.0	27.3	1.19	0.73	370	191	179	9
SAG 1.5B	68.4	54.4	13.6	20.3	1.26	0.67	339	212	127	9
SAG 3.2B	67.1	55.2	45.8	69.7	1.21	0.66	608	204	404	9
SAG 6.6B	70.4	55.7	25.2	44.5	1.26	0.56	391	142	249	10
SAG 9.4B	82.9	67.7	5.3	7.1	1.22	0.75	255	180	75	14
SAG 10.3C	66.7	54.2	28.0	41.8	1.23	0.67	521	209	312	11
SAG 11.1C	73.3	61.5	30.9	43.8	1.19	0.71	626	260	366	12
TUR 1.5B	71.4	58.5	10.7	16.0	1.22	0.67	293	181	112	10
TUR 2.3B	68.5	56.7	10.4	14.9	1.21	0.70	311	208	103	10
TUR 3.2B	92.5	79.4	19.5	26.3	1.17	0.74	366	196	170	9
TUR 4.6B	68.7	57.6	19.2	27.1	1.19	0.71	318	198	120	6
TUR 5.2B	83.3	70.0	5.8	8.0	1.19	0.72	294	192	102	18
TUR 7.5B	66.3	52.2	7.4	12.1	1.27	0.61	269	198	71	10
Statistics										
Mean	71.0	58.5	19.3	28.4	1.22	0.69	387	197	190	11
S.d.	7.5	7.1	11.1	16.9	0.03	0.04	115	25	103	3
S.d./mean (%)	11	12	58	59	2	6	30	12	54	25

Parameters: $H_{c(r)}$ = (remanent) coercivity force (in mT); $J_{(r)s}$ = (remanent) saturation magnetisation (in A/m) $(A) = H_{CR}/H_C$ ratio; $(B) = J_{RS}/J_S$ ratio, X_{LF} = low-field susceptibility (SI-units); X_{HF} = high-field susceptibility (SI-units); d ($\times 10^{-6}$ SI-units) is the difference between the low- and high-field susceptibility ($d = X_{LF} - X_{HF}$); and $(C) = d$ to J_{RS} ratio (in mm/A).

a fresh site (SAG 11.1B) shows a complicated thermomagnetic behaviour (Fig. 13). The sample has an initial NRM intensity of 29.6 mA/m; another sample from this same site shows a major drop in X_{LF} between 150 and 320°C (Fig. 12). Its thermomagnetic behaviour (heating and cooling curves) was recorded during three runs of heating to successively higher temperatures (300, 450 and 600°C) and cooling to room temperature (Fig. 13: A, B, C). There is a major decrease in magnetisation between 250 and 430°C; the cooling curves from 300 and 450°C (Fig. 13: A, B) do not return along the same trajectory as the heating curves. The irreversibility of the curve shows that the maximum blocking temperature of 430°C is not a Curie temperature; the irreversibility is likely caused by chemical alterations during heating. The curves from the weathered samples do not show this decrease; chemical alterations had occurred already in the outcrop. For all samples a minor increase in magnetisation occurs between 430 and 500°C (Fig. 13) indicating the formation of magnetite. The hyperbolic shape of the cooling curves is strongly dominated by the paramagnetic clay minerals.

Hysteresis measurements

Hysteresis curves were determined by measuring the imbalance of a pair of coils, one of which contained the specimen. Measurements were carried out using an alternating field with a maximum of 250 mT. All curves show in high fields a linear magnetisation/field behaviour, which is caused by the paramagnetic contribution of the clay minerals. The slope of this linear increase at high fields denotes the high field susceptibility (X_{HF}) (Fig. 14; Table 4).

Hysteresis measurements were performed for 35 samples from the collection. The coercive force

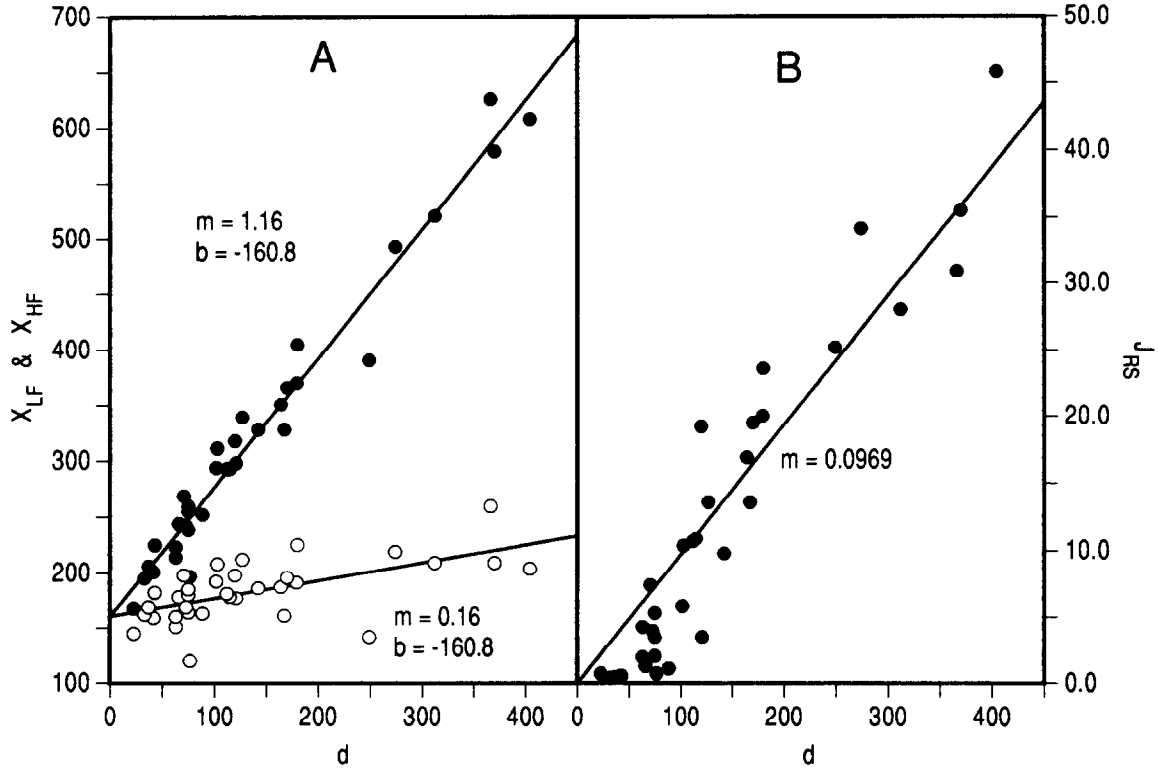


Fig. 15. Some rock magnetic parameters (X_{LF}, X_{HF} and J_{RS}) in relation to the difference d ($= X_{LF} - X_{HF}$) which is a measure for the amount of remanence carrying minerals. (A) X_{LF} (dots) and X_{HF} (circles) in relation to "d"; regression line gives an estimate for the paramagnetically controlled "base-level" for the clays: 161×10^{-6} SI-units. (B) Relation between J_{RS} and "d"; regression line gives d/J_{RS} ratio (10.3 mm/A) which is independent on the quantity of ferrimagnetic minerals and gives an indication of constant grainsize for a constant mineralogy in these clays.

(H_C), saturation magnetisation (J_S) and remanent saturation magnetisation (J_{RS}), were determined from hysteresis measurements for samples from nearly all sites using the method of M. Jackson et al. (1990). For the determination of the remanent coercivity (H_{RC}) of the ferrimagnetic minerals, the paramagnetic contribution is subtracted (Fig. 14; Table 4). The results of samples having a J_{RS} below 5 A/m are shown in different Tables because of the large inaccuracy in the calculated rock magnetic parameters (J_S, H_{CR} and H_C). Only the samples with J_{RS} above 5 A/m were used for the characterisation of the magnetic mineralogy of these clays. In fact, the samples with J_{RS} above 5 A/m are clearly the fresh samples, sometimes having a small secondary LT component; the other samples (J_{RS} < 5 A/m) have less accurate HT directions or no HT magnetisations at all.

The X_{HF} is rather constant for all samples (Table 4); its value is $(150–200) \times 10^{-6}$, which concurs with the earlier suggestion (from the X_{LF} versus temperature plot; Fig. 12) of a paramagnetically controlled "base-level" for the X_{LF} of these Pleistocene clays. The difference (d) between X_{LF} and X_{HF} (Table 4) is a measure for the amount of remanence carrying (ferrimagnetic) minerals. There is a linear relation between X_{LF} and d (Fig 15A) because of the constant X_{HF} values; the regression line through the data points implies a base-level for the X_{LF} of 161×10^{-6} SI-units. For X_{HF} there is not a perfect horizontal line (Fig. 15A) as expected for a constant X_{HF} value; this implies that either a part of the X_{HF} is determined by a non-oxidized paramagnetic minerals (for example pyrite) or that the hysteresis loop is not fully saturated at high fields for the samples having a high J_{RS} due to for example the presency of goethite. We prefer the first option, implying the presency of non-oxidized pyrite in the fresh sites; pyrite is a common mineral in sediments.

The vertical separation between the descending branches of a hysteresis loop, as a function of the applied field, is a measure of the sample's (cumulative) remanent coercivity spectrum (M. Jackson et al., 1990). This spectrum was determined from hysteresis measurements from the

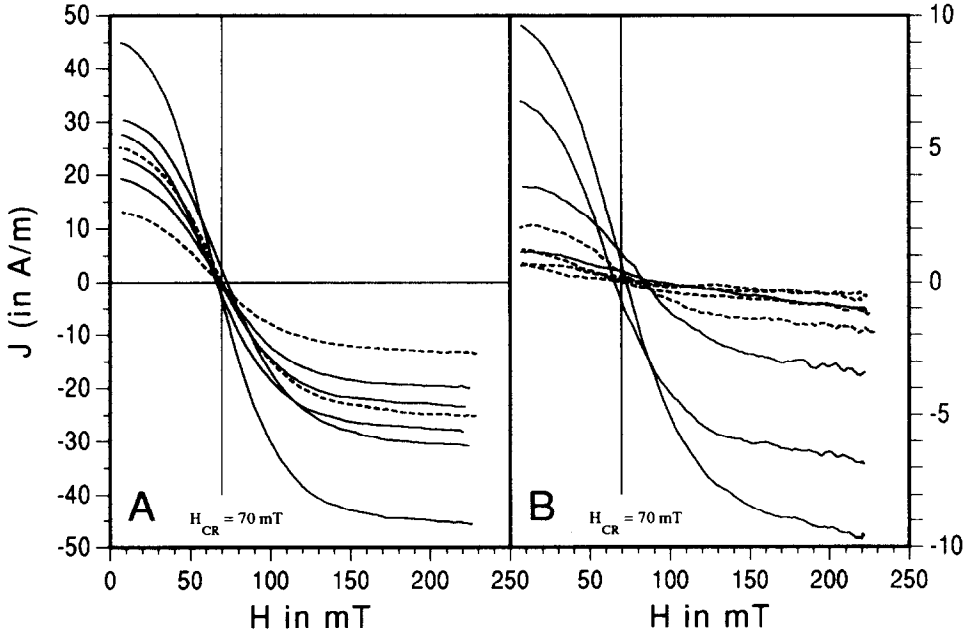


Fig. 16. Collection of calculated remanent coercivity spectra of the studied clays from the Sant'Arcangelo basin (SAG) and Pisticci (PST) as deduced from hysteresis measurements. Solid lines give the samples with a primary LT component; dotted lines the samples with a secondary LT component (affected by weathering). (A) Samples for which $J_S > 10$. (B) Samples for which $J_S < 10$. Intersection with horizontal line $J = 0$ gives the remanent coercivity force (H_{CR}); all samples (fresh and weathered) have a H_{CR} of 70 A/m. Differences in (B) are due to inaccuracy.

Sant'Arcangelo–Pisticci sites (Fig. 16); it serves as an illustration of the hysteresis measurements and the reliability of the method of M. Jackson et al. (1990). The shape of the individual curves is nearly identical (i.e. constant H_{RC} values). but the differences in absolute values of J_{RS} are very large. Probably, the mineral assemblage carrying the NRM is the same for the different degrees of weathering; the sediment loses its remanence without creating new remanence carrying minerals disturbing the original NRM; the H_{RC} of this mineral is around 70 mT, between 60.4 and 92.5 mT (Table 4); the average H_{RC} value for the samples with $J_{RS} > 5$ A/m is 71.0 ($\pm$7.5) mT.

The remanent saturation magnetisation (J_{RS}) is increasing linearly with "*d*" (Fig. 15B). As expected, the J_{RS} depends on the concentration of ferrimagnetic minerals. The ratio of J_{RS} to J_S is high and very constant for these clays (0.69 $\pm$ 0.04; Table 4). This ratio exhibits a maximum for single-domain magnetite grains (Day et al., 1976). A magneto-crystalline model for pyrrhotite assuming a unique hard axis and a triaxial anisotropy in the unique plane was used to calculate a J_{RS}/J_S ratio of 0.750 (O'Reilly, 1984; after Dunlop, 1971). Therefore, we believe that an iron sulphide mineral (greigite or pyrrhotite) carries the main part of the remanence of these sediments. The ratio of the H_{RC} to H_C increases with increase in grain size; for these clays the value is 1.22 ($\pm$0.03, Table 4), implying 0.1 mm grains (O'Reilly, 1984; after Day et al., 1976). The iron sulphide mineral carrying the main part of the remanence is not the mineral responsible for the characteristic (HT) component, with maximum blocking temperatures above 320–340°C. The rock magnetic properties of the mineral carrying the HT magnetisation could not be determined because this mineral carries only a minor part of the remanence.

Magnetic minerals carrying the NRM of these clays

Our rock magnetic observations support the idea that the remanence of these clays is carried by two magnetic minerals. Typically, the fresh sites show an iron sulphide magnetisation removed at 320–340°C, this mineral is either pyrrhotite or greigite. Greigite occurs in sediments as a result of microbial activity and is reported to be magnetic (Tarling, 1983). The fresh sites show a strong decay in susceptibility and NRM intensity during removal of this iron sulphide component. The presence of iron sulphides carrying the LT magnetisation is supported by the large J_{RS}/J_S ratio. Indications of pyrrhotite have been found earlier in Pliocene marls about 400 km south of the Matera area in Calabria (Linssen, 1988). Sulphate reduction to pyrite below the iron-reduction interval in suboxic to anoxic sediments may have produced authigenic minerals like greigite and possibly pyrrhotite. Removal (i.e. low-temperature oxidation or weathering) of the authigenic minerals in the outcrop caused a considerable loss in X_{LF} to a para-magnetically controlled "base-level" of 160×10^{-6}. The iron sulphides do not carry the characteristic (HT) component of these clays, but they carry in principle (one of) the primary magnetisations. Sites with iron sulphide magnetisations with opposite polarity relative to the HT component have very discriminating properties concerning NRM directions (for example Fig. 5B, C). Thermomagnetic measurements show that the removal of this iron sulphide magnetisation is mainly due to chemical alterations and not (only) because of reaching the Curie temperature.

For the other magnetic mineral-carrying the HT magnetisation—a maximum blocking temperature of 430–480°C was observed. Nearly all sites from the Matera area and the Apulia–Gargano foreland have the (HT) component removed between 360 and 480°C. It is clearly acquired before tectonic tilting of the sediments and therefore most likely of primary origin. The nature of this mineral is very speculative; probably it is a titanomagnetite ($Fe_{3-x}\ Ti_x\ O_4$) of detrital origin introduced by the major volcanic activity in this region; according to the maximum blocking temperature of 430–480°C (see Tarling, 1983): $0.2 < x < 0.25$. In marine marls of Pliocene age more to the south (Calabria and Sicily), blocking temperatures are usually much higher (580–600°C, Langereis and Hilgen, 1991) and indicate clearly the presence of single domain magnetite (Van Velzen and Zijderveld, 1990) which may be slightly cation deficient (Linssen, 1988; 1991).

Discussion and conclusions

In general, two components (and mineral types) have contributed to the NRM of the studied late Tertiary and Quaternary sediments from the Matera area. A low-temperature (LT) component is removed between 100 and 300–320°C and it is carried by iron sulphides; a high-temperature (HT) component is removed between 300–320 and 430–450°C and is probably carried by titanomagnetites. From the clays of the Sant'Arcangelo basin and the Pisticci section, there is evidence that (even) moderate weathering replaced a high-intensity primary (often reversed) LT component with a low-intensity secondary (normal polarity) LT magnetisation. The HT magnetisation is affected by this process by a decrease in intensity but not by a change in direction, although sometimes extensive weathering does affect the HT magnetisation (site PST 8–9) or may even remove it completely (site PST 2).

Timing and amount of rotation

The sites from the Matera area give consistent paleomagnetic results; a positive fold test of the HT component supports its primary (= pre-folding) origin. For the sites in the Apenninic belt, declinations show on average a 23° counterclockwise rotation (Table 5) relative to the geographic North. The sediments from these areas have ages between 2.02 and 0.70 Ma (see Appendix) and hence all reversed polarities must represent the Matuyama Chron. The normal polarity sites in the Craco area represent most likely the Olduvai Subchron.

For the three localities in the Bradano Unit, declinations show on average no rotation (Fig. 10). Here, reversed polarities also belong to the Matuyama Chron; but considering their age (0.90 Ma) the normal polarity sites in the Pisticci section most probably represent the Jaramillo Subchron (middle part section) and the Brunhes Chron (upper part section). The earlier sampled

TABLE 5

Earlier paleomagnetic results from the southern Apennines and the Apulian foreland and the comparison to the results from this study

N	*D*	*I*	*k*	Locality	Age	Reference
Apulian Unit						
4 sites	324.4	36.0	480	Gargano peninsula	Cretaceous	Channell and Tarling (1975)
17 sites	335.0	37.8	31	Gargano peninsula	Late Cretaceous	Channell (1977)
21 sites	327.7	38.2	56	Gargano peninsula	Late Cretaceous	VandenBerg (1983)
3 studies	329.0	37.4	328	Gargano peninsula	Cretaceous	VandenBerg (1983)
5 areas	359.8	56.3	491	Apulian Unit	early Pleistocene	Scheepers (1992)
Bradano Unit						
3 areas	359.1	56.3	218	Matera area	early Pleistocene	this study
Southern Apennines, internal part						
3 sites	285.8	54.1	48	Capri (CL platform)	Senonian (Late Cretaceous)	Catalano et al. (1976)
5 sites	286.6	44.5	21	Mt.Cerviero (CL platform)	Late Cretaceous	Manzoni (1975)
113 samples	248.0	30.4	48	Pescopagnano (LN basin)	Cretaceous	Incoronato (1983)
Southern Apennines, external part						
5 sites	322.4	28.2	120	Campia (Matese Mt.)	Cretaceous	Channell and Tarling (1975)
1 site	322.2	30.9		Maggiore (AC platform)	Senonian (Late Cretaceous)	Catalano et al. (1976)
9 sites	326.0	42.0	44	Monte Maiella	Cretaceous	K.C. Jackson (1990)
19 sites (*)	312.0	61.0	50	Monte Raparo	Cretaceous (remagnetization?)	K.C. Jackson (1990)
3 areas	271.1	44.6	16	Internal part	Cretaceous	K.C. Jackson (1990)
3 areas	323.4	33.7	116	External part	Cretaceous	K.C. Jackson (1990)
13 sites	337.8	56.1	111	Sant'Arcangelo basin	early Pleistocene	Sagnotti (1992)
3 areas	336.6	57.2	254	Matera area	early Pleistocene	this study

Symbols the same as used in Table 3, further: CL platform = Campania–Lucana platform; LN basin = Lagonegro basin; AC platform = Abruzzi–Campania platform.

sites on the Apulia–Gargano foreland also showed no rotation (Scheepers, 1992). Thus, it follows that only the Apenninic belt underwent a counter-clockwise rotation of 23° in the Pleistocene. According to the age estimates of the sites, this rotation was at least younger than 1.35 Ma and hence took clearly place after the major phase of extension related to the formation of oceanic crust, between 2.0 and 1.6 Ma, in the southeastern part of the Tyrrhenian basin (Sartori et al., 1989).

Relation to neotectonic evolution

The geodynamical evolution of the southern Italian area since the late Pliocene is dominated by the extension related to the opening of the southern Tyrrhenian basin (between 2.0 and 1.6 Ma). The extension is characterised by radial and concentric fault systems. In the southern Apennines, tectonic activity and sedimentation shifted eastwards and southwards during the Neogene and Quaternary. Passive subduction of a lithosphere slab by gravitational sinking has been regarded as the mechanism of this process inducing tectonic block rotations. But from this study it can be concluded that these block rotations occurred later than the major phase of extension.

In the early Quaternary a phase of compression known as the "Quaternary compressive phase" temporarily replaced extension. It took place between 1.0 and 0.70 Ma (Philip and Tortorici, 1980; Bousquet and Philip, 1981; Auroux et al., 1986). This phase was responsible for the activation of numerous reverse faults and associated folds in the Bradano Unit. Compression related to these structures is NE–SW directed. In the Apulia–Gargano foreland, minor strike slip faulting took place in this period, also associated with a NE–SW shortening direction (Bousquet and Philip, 1986). This study shows that probably during this "Quaternary compressive phase" major tectonic rotations also occurred.

Extent of block rotations

The mean direction for Cretaceous rocks from the Apulia–Gargano foreland implies a 31° counter-clockwise rotation (Table 5) during the Tertiary. The paleomagnetic data of Cretaceous rocks from the southern Apenninic belt (Channell and Tarling, 1975; Catalano et al., 1976; Incoronato, 1983; K.C. Jackson, 1990) are quite scattered, nevertheless they all imply counter-clockwise rotations between 37.6 and 89° (Table 5). Cretaceous limestones at Monte Raparo some 20 km west of the Matera area show a 48° counter-clockwise rotation (K.C. Jackson, 1990); because of steep inclination values the magnetisation in these Monte Raparo limestones was probably acquired in connection with the Apenninic deformation. The sites from the internal part of the Apenninic belt show more rotation (89°) than the ones from the external part (Table 5): the three areas in Mesozoic rocks from this part of the Southern Apennines only reveal a 37° counter-clockwise rotation.

The results from this study give slightly smaller amounts (23°) of rotation for early Pleistocene sediments from the western part of the Bradano Unit. Although the differential rotations between Mesozoic rocks in the southern Apennines are probably due to rotational movements of thrust sheets, we suggest that the whole area of the southern Apennines behaved as one single coherent block during this middle–late Pleistocene rotation phase. The idea of a single block is also supported by the seismicity (Gasparini et al., 1982) and micro-tectonic analysis (Auroux et al., 1986). Micro-tectonic analysis shows that the direction of the "Quaternary compressive phase" is homogenous for the whole area (Auroux et al., 1986).

Acknowledgements

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reviews. We are grateful that the tectonic system of the southern Apennines remained quiet during the sampling period.

Appendix

In order to secure a more accurate age assessment than the general late Pliocene–early Pleistocene age of the sites sampled in the Matera area, we qualitatively analysed the foraminiferal content in the (sieved) size fraction > 125 μm. From a biostratigraphic and biochronologic point of view, the planktonic foraminiferal taxa *Globorotalia inflata*, *Globorotalia crassaformis*, *Globorotalia truncatulinoides* and sinistrally coiled neogloboquadrinids, and the benthic species *Hyalinea balthica* and *Islandiella ionica* are relevant (Table A1).

The presence/absence of these foraminiferal marker species are presented in Table A2. In combination with the stratigraphical order, which is well known for the TUR and

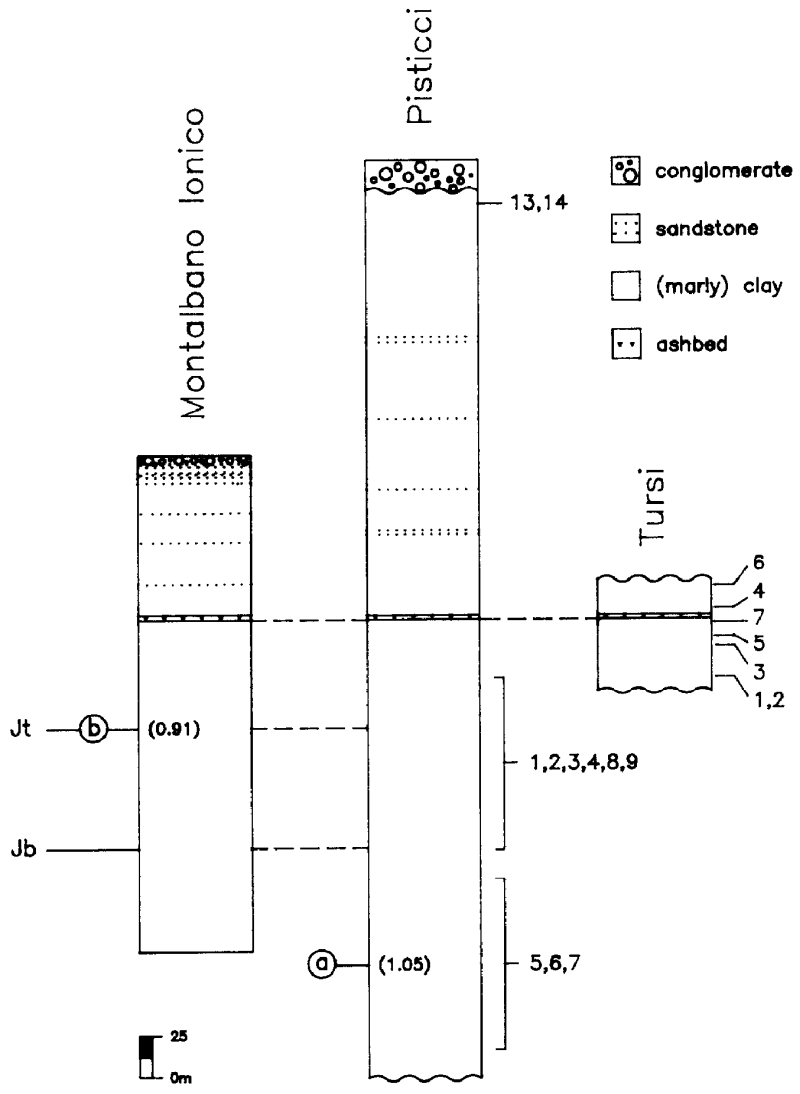


Fig. A1. Lithology and position of samples in the Pisticci and Tursi sections. Lithologic column of the Pisticci section after Lentini (1971). The position of the PST samples with respect to Lentini's log is only known by approximation. The position of the re-appearance level of *G. truncatulinoides* (*a*) in this section — with an estimated age of 1.05 Ma — is also based on Lentini (1971). The Montalbano Ionico section of Verhallen (1990) has been added because this section provides an additional useful biostratigraphic datum plane, i.e. the top of the small *Gephyrocapsa* Zone (*b*) with an estimated age of 0.91 Ma (Rio, 1982). The three sections have been correlated lithostratigraphically on the basis of the very distinct ashbed. Finally, the estimated positions of the top and bottom of the Jaramillo (*Jt* and *Jb*) have been indicated, based on linear interpolation between the two biostratigraphic datum planes (*a* and *b*) and assuming that the sedimentation rate is constant and the same in all three sections.

TABLE A1

Foraminiferal datum planes used to determine the age of the samples by means of biostratigraphic analyses

Foraminiferal species	Event	Age (Ma)
Globorotalia truncatulinoides	re-appearance	1.05
Islandiëlla ionica	last occurrence	1.35
Hyalinea balthica	first occurrence	1.45
Globorotalia crassaformis	influx	1.60
Sinistral neogloboquadrinids	first common occurrence	1.68
Islandiëlla ionica	first occurrence	1.83
Globorotalia inflata	re-appearance	1.93
Globorotalia truncatulinoides	first occurrence (influx)	1.95
Globorotalia inflata	first occurrence (influx)	2.02

Ages of most datum planes have been obtained by linear interpolation or extrapolation of sediment accumulation rates in the Vrica section (Zijderveld et al., 1991), using conventionally dated magnetic reversal boundaries as calibration points (ages after Berggren et al., 1985). Ages of benthic foraminiferal datum planes are based on the stratigraphic positions of these levels in the Vrica section after Selli et al. (1977) and Verhallen (1990). The age of the re-appearance level of *G. truncatulinoides* is taken from Rio et al. (1991; fig. 6).

PST sites (Fig. A1) but unknown for the SAG sites, these data provide us with initial age estimates for the samples. The presence of *H. balthica* and dominantly sinistrally coiled assemblages of neogloboquadrinids indicate that all PST and TUR samples most probably have an age younger than 1.45 Ma. This maximum age applies to several of the SAG samples as well, but a slightly older age—2.02 Ma at a maximum—cannot be excluded for a number of sites (i.e. SAG 1, 4, 7, 8 and 9; see Table A2). On the other hand, the absence of *H. balthica* at all CRC sites provides strong negative evidence that these sites are older than 1.45 Ma whereas the presence of *G. inflata* in most samples indicates that they are younger than 2.02 Ma. This is consistent with the presence of *G. ionica* at CRC 4 indicating an age between 1.83 and 1.35 Ma and of *G. crassaformis* at CRC 1. The excellent preservation of the latter species and its absence at all other sites from the Matera area seem to exclude reworking from older sediments. As a consequence, the level of CRC 1 should correspond to the early Pleistocene influx of *G. crassaformis*, which directly succeeds the top of the Olduvai in both the Singa and Vrica sections in Calabria, dated at ± 1.60 Ma (Zijderveld et al., 1991). At the same this implies that the normal polarity site CRC 3 most probably represents the Olduvai.

The absence of *G. truncatulinoides* in all samples but one, on the other hand, does not yield a minimum age estimate for our samples because of the very rare and inter-mittent occurrence of this species in the Mediterranean directly after its re-appearance around 1.05 Ma. Although such a minimum age estimate cannot be obtained on the basis of the present foraminiferal dataset, the dominantly (overall) reversed polarity clearly suggests that all samples are older than the Brunhes/Matuyama boundary dated radio-metrically at 0.73 Ma (Mankinen and Dalrymple, 1979). Existing data from the literature can be used, however, to date the TUR and PST sites more accurately.

The general stratigraphy of the Plio-Pleistocene succession in the area has been subject of a study by Lentini (1967). In a subsequent paper, the same author describes the Pisticci section in considerable detail (Lentini, 1971). The Montalbano Ionico section, located between Pisticci and Tursi (Fig. 2) and not used in the present study because of the rather intense weathering, has been dealt with by Verhallen (1990). A distinct ash layer provides a very reliable lithostratigraphic correlation between the different sections (Fig. A1). Especially relevant are the position of the top of the small *Gephyrocapsa* Zone, stratigraphically 65 m below the ash layer in the Montalbano Ionico section (Verhallen, 1990; see also Fig. A1), and the occurrence of *G. truncatulinoides* well above the entry level of *H. balthica* and associated with dominantly sinistrally coiled neogloboquadrinids at a distinctly lower stratigraphic level, some 200 m below the ash, in the Pisticci section (Lentini, 1971). The re-appearance of large-sized *Gephyrocapsa* (= top of the small *Gephyrocapsa* Zone) represents an important biostratigraphic datum plane which has been dated at 0.91 Ma and is practically coincident with the top of the Jaramillo (Rio, 1982). Unfortunately, the base of the small *Gephyrocapsa* Zone was not reached at Montalbano Ionico (Verhallen, 1990). The *G. truncatulinoides*-bearing level in the Pisticci section most likely corresponds to the re-appearance level of this species in the Mediterranean dated at 1.05 Ma. If we use these two biostratigraphic datum planes as age calibration points and assume that the sedimentation rate was equal in the nearby Pisticci and Montalbano Ionico sections and remained constant through time, linear extrapolation yields an age of 0.84 Ma for the ash layer. These ages have to be considered however as minimum ages because the sedimentation rate can be assumed to have increased in view of the shallowing upward trend in the succession (see Verhallen, 1990). Linear interpolation further yields an approximate position for the lower Jaramillo boundary in our sections. Application of our simple age model further implies that the ages of the TUR sites range from 0.78 to 0.88 Ma. They postdate the top of the Jaramillo and predate the Brunhes/Matuyama boundary. This is consistent with the observed reversed polarity of these samples (see section "Paleomagnetic results of the Matera area"). Although the stratigraphic distance of the Pisticci sites to the ash layer has not been measured, the position of these sites with respect to the lithologic log of Lentini (1971) was determined, with an accuracy of less than 50 m. The lower sites PST 5–7 correspond approximately to Lentini's sample level 25 and the middle sites PST 1–4 and 8–9 to sample level 30 (see Fig. A1). The upper sites PST 13–14 were taken just below the

TABLE A2

Bedding orientation, polarity, biostratigraphy and ages of the sampled sites

Site	Strike	Dip	Pol	A	B	C	D	E	F	Age (Ma)
TUR 6	342	10	R	–	x	?	–	?	–	0.82
TUR 4	317	6	R	–	x	1:09	–	x	–	0.84
TUR 7	195	7	R	–	x	?	–	?	–	0.84
TUR 5	317	6	R	–	x	0:06	–	x	–	0.85
TUR 3	317	6	R	–	–	0:01	–	x	–	0.86
TUR 2	317	6	R	–	x	1:09	–	x	–	0.89
TUR 1	317	6	R	–	x	1:09	–	x	–	0.89
PST 13	315	7	N	–	–	?	–	x	–	< 0.73
PST 14	315	7	N	–	x	0:06	–	x	–	< 0.73
PST 9	315	7	R	–	x	?	–	?	–	0.88
PST 8	315	7	R	–	x	?	–	?	–	0.89
PST 3	315	7	R	–	–	8:02	–	x	–	0.90
PST 1	315	7	N	–	x	9:00	–	x	–	0.92
PST 2	315	7	?	–	x	7:03	–	x	–	0.93
PST 4	315	7	N	–	x	?	–	x	–	0.94
PST 7	nc	nc	R	–	x	0:02	–	x	–	± 1.05
PST 6	nc	nc	R	–	x	2:09	–	x	–	± 1.05
PST 5	nc	nc	R	–	x	7:03	–	x	–	± 1.05
POM 1	0	0	N	–	x	?	–	x	–	< 0.99
POM 2	0	0	N	–	x	0:04	x	x	–	< 0.99
POM 3	116	7	N	–	x	0:07	–	x	–	< 0.99
SAG 1	265	36	R	–	x	0:09	–	–	–	< 2.02
SAG 2	180	42	R	–	x	0:09	–	x	–	< 1.45
SAG 3	180	20	R	–	x	?	–	x	–	< 1.45
SAG 4	300	30	R	–	x	2:08	–	–	–	< 2.02
SAG 5	230	20	R	–	x	0:05	–	x	–	< 1.45
SAG 6	270	38	R	–	x	9:03	–	–	–	< 1.68
SAG 7	140	25	R	–	–	1:09	–	–	–	< 2.02
SAG 8	140	25	R	–	x	1:09	–	–	–	< 2.02
SAG 9	142	30	R	–	x	2:08	–	–	–	< 2.02
SAG 10	118	18	R	–	x	7:03	–	–	–	< 1.68
SAG 11	106	24	R	–	x	9:05	–	–	–	< 1.68
CRC 1	295	24	R	x	x	1:09	–	–	–	1.60
CRC 2	295	24	R	–	–	0:02	–	–	–	?
CRC 3	273	22	N	–	x	1:09	–	–	–	1.45–2.02
CRC 4	126	21	R	–	x	0:10	–	–	x	1.45–1.83
CRC 5	272	39	N	–	x	0:10	–	–	–	1.45–2.02
CRC 6	170	25	R	–	–	1:09	–	–	–	?

nc = not clear.

PST and TUR samples are given in stratigraphical order. Polarity of samples (pol) is based on the results of thermal demagnetization. Biostratigraphic data mark the presence/absence of the foraminiferal marker species *G. crassaformis* (A), *G. inflata* (B), G. *truncatulinoides* (D), *H. balthica* (E) and *I. ionica* (F), and the ratio of sinistrally and dextrally coiled neogloboquadrinids (C). Ages of TUR samples have been calculated by linear extrapolation of the sediment accumulation rate using the top of the small *Gephyrocapsa* Zone—dated at 0.91 Ma (Rio, 1982)—in the Montalbano Ionico section and the re-appearance level of *G. truncatulinoides*—dated at 1.05 Ma—in the Pisticci section, and assuming that the sediment accumulation rate is constant and the same in all three sections. Ages of PST samples 1–4, 8 and 9 are based on their relative stratigraphic position (with an approximate sample spacing of 10 m), a sediment accumulation rate of 96 cm/kyr and the conclusion that the recorded polarity reversal represents the top of the Jaramillo dated at 0.91 Ma (Berggren et al., 1985). PST samples 5–7 were collected from the lower part of the regular section and correspond approximately with sample level 30 of Lentini (1971) dated biostratigraphically at 1.05 Ma. Ages of SAG samples are based on the (foraminiferal) biostratigraphic data (Table A1) because the stratigraphic position of these samples with respect to the ashbed is not known. The ages for the SAG sites are maximum ages because of the less definitive character of these biostratigraphic datum planes in case the marker species is absent. The normal polarity of the POM sites has been used to further constrain the age of these sites.

marine terrace and corresponds approximately to sample levels 44/45 of Lentini (1971).

These approximate positions indicate that the lower sites (PST 5–7) can be dated around 1.05 Ma and are older than the Jaramillo. This agrees with the observed reversed polarity of the samples (see section "Paleomagnetic results of the Matera area"). On the other hand, the middle sites fall either within the Jaramillo or are positioned slightly on top of it. Finally, the normal polarity upper sites (PST 13–14) indicate that the Brunhes is reached. This conclusion is consistent with linear extrapolation of the sedimentation rate. Considering the polarity and stratigraphic order, the normal sites PST 1 and 4 belong to the Jaramillo and the remaining PST sites, located slightly higher in the section (with a spacing of some 10 m), to the reversed interval directly postdating the Jaramillo. The ages of the SAG sites cannot be further constrained.

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